CCH® Tagetik | **ACADEMY**

Planning and forecasting



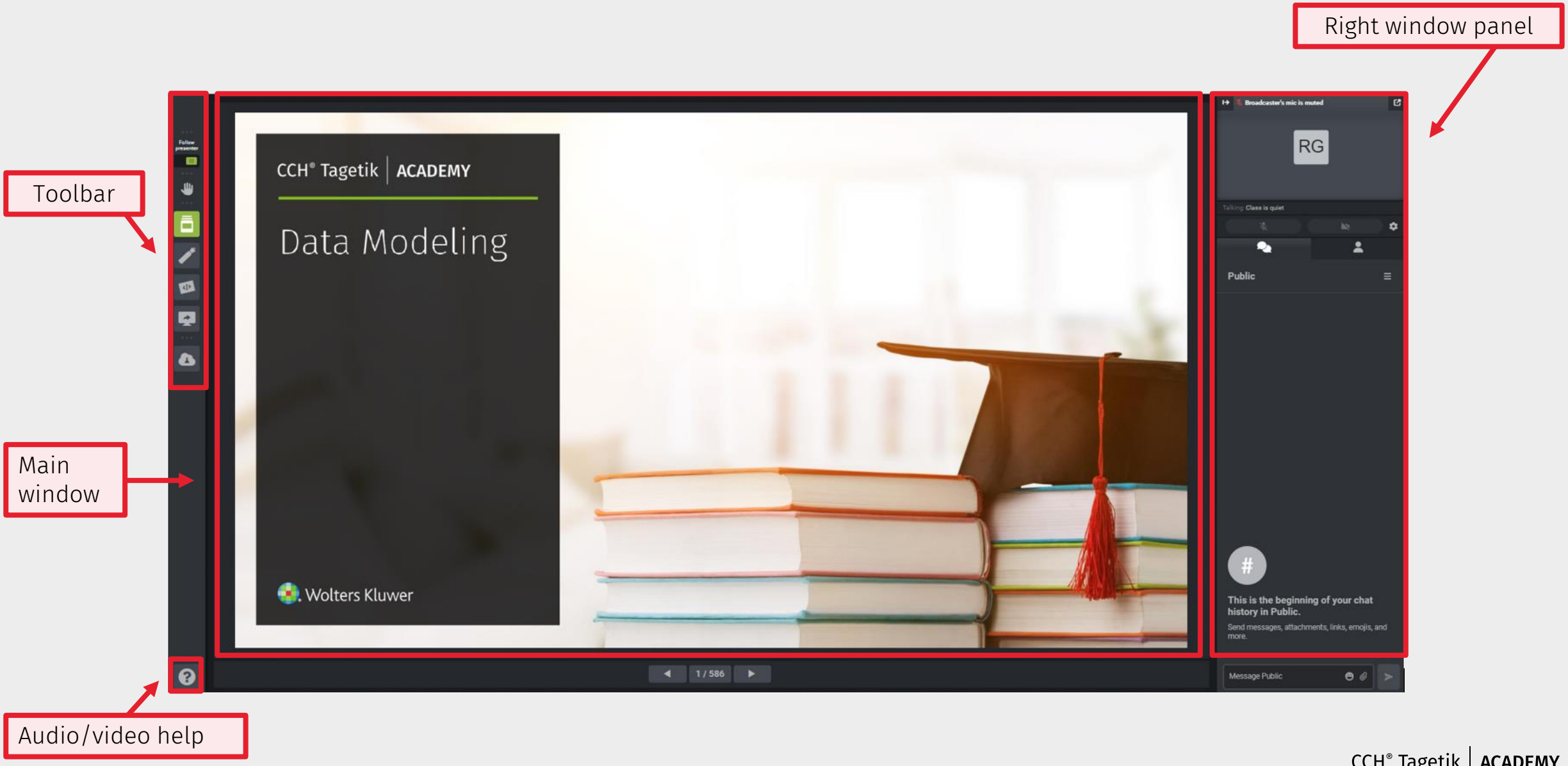
Wolters Kluwer

Before starting

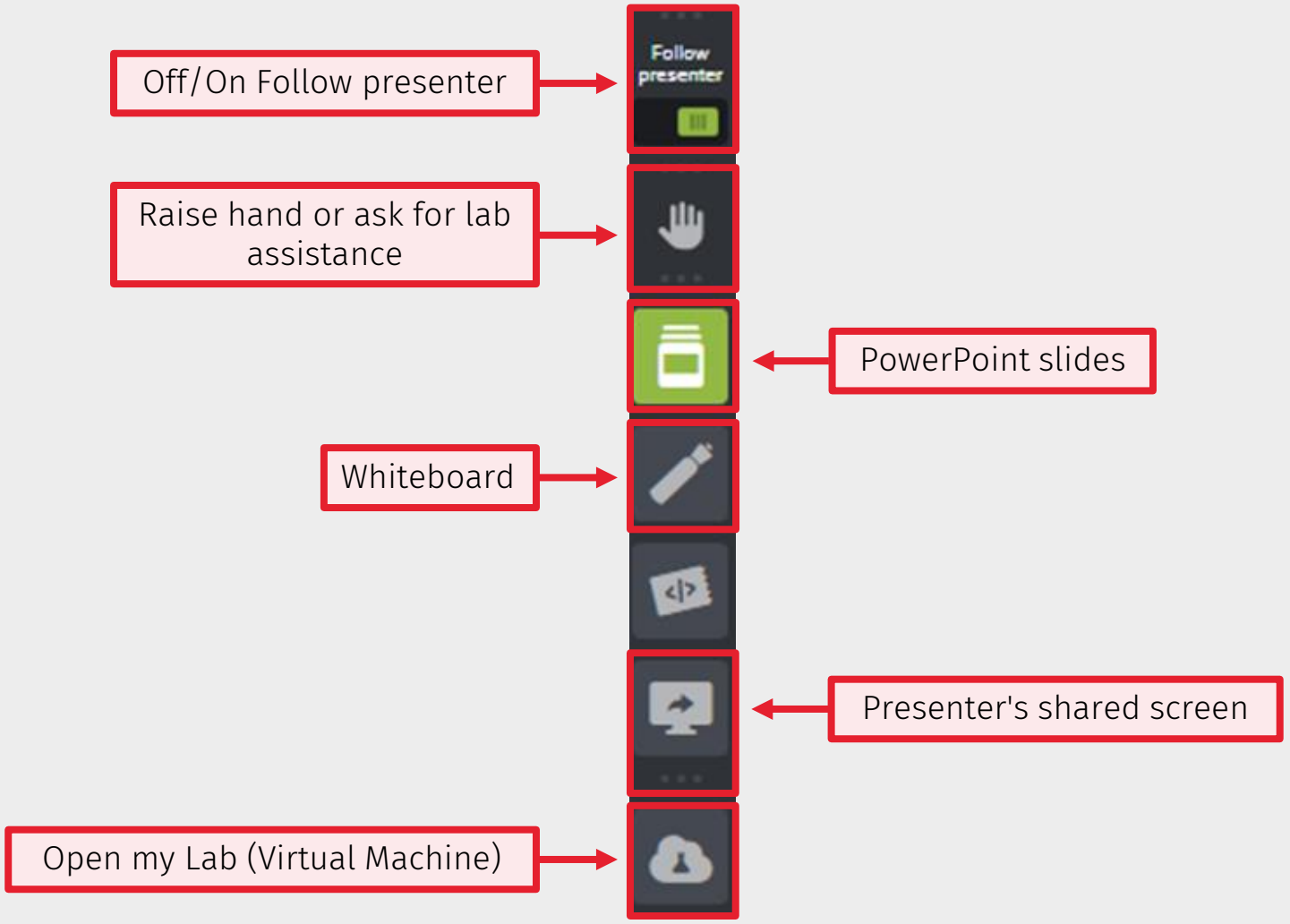
Strigo platform - “How to”



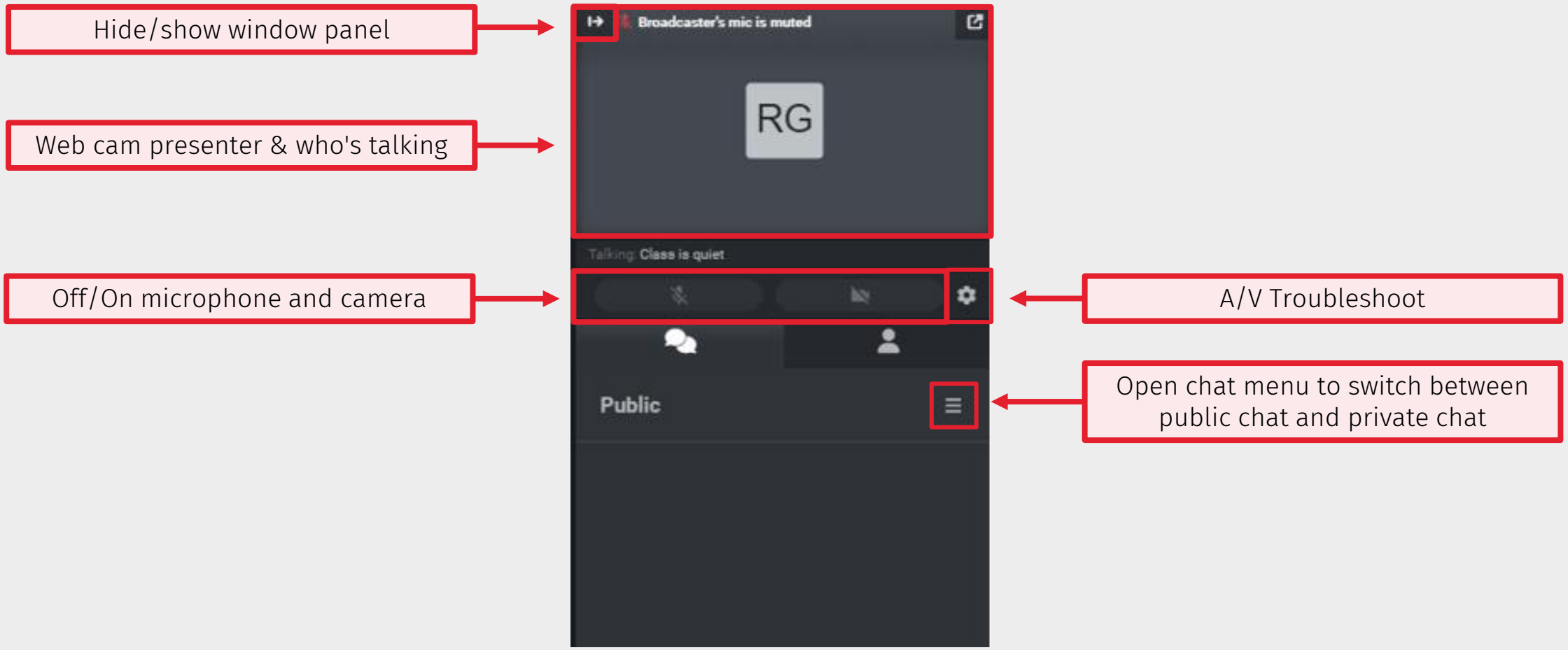
Strigo overview (for Instructor-led classes)



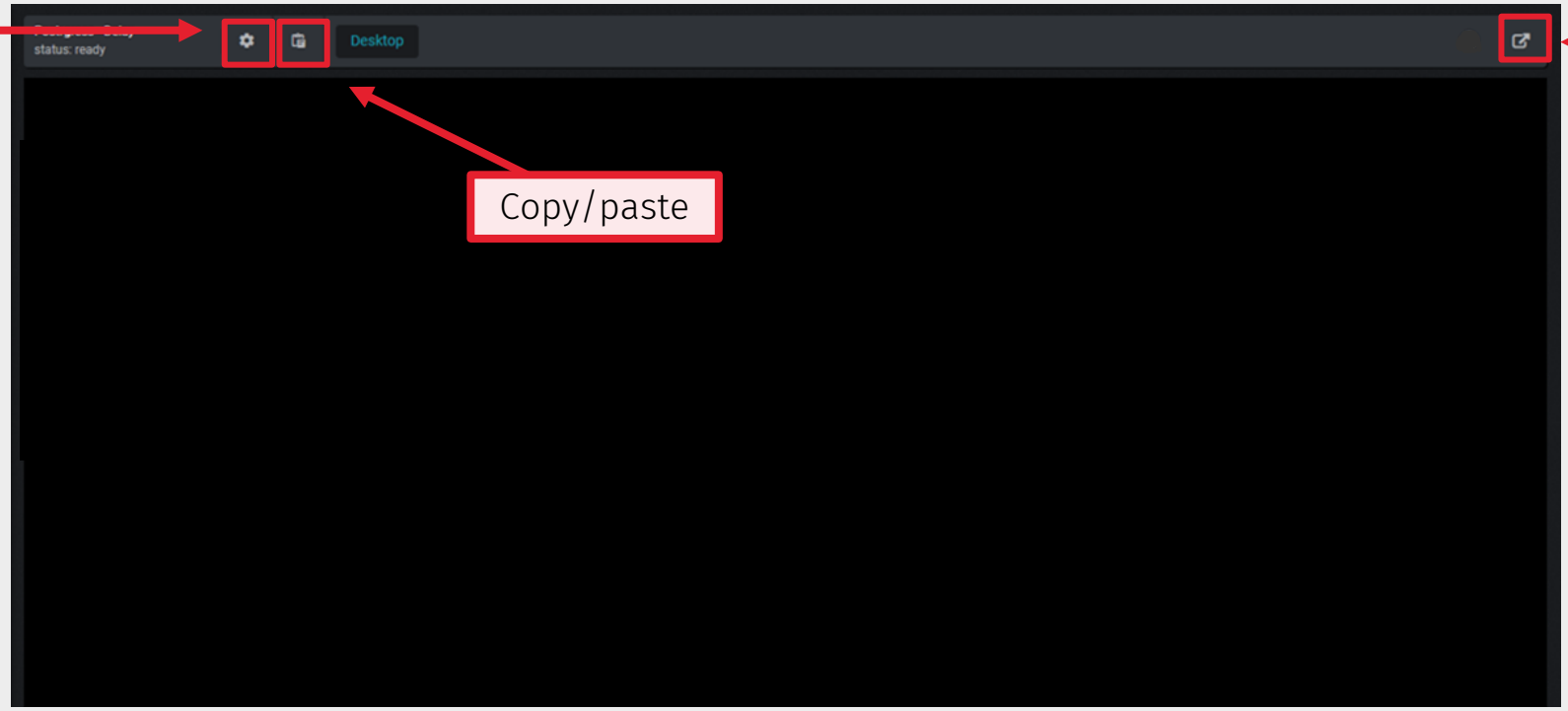
Strigo left toolbar



Strigo right window panel



Strigo Lab (Virtual Machine)

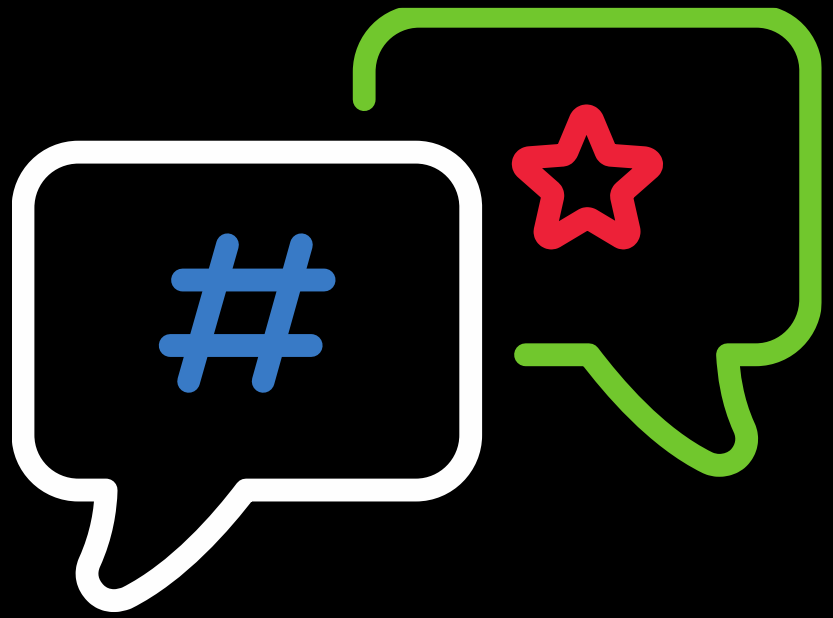


Agenda

- Introduction
- Data model trends
- Top-down planning
- Bottom-up planning
- Driver based planning
- Assumptions
- Reporting
- Versioning
- Forecasting



Let’s talk about: Planning and forecasting introduction



Planning – the what

Determinate or **estimate** over a specific period of time with the overall goal to **optimize the future performance** of an organization establishing **objectives and actions**

Planning – the why

Future

- Budgeting helps in anticipating for future uncertainties and minimizing risks
- It provides a criterion to measure actual results against goals

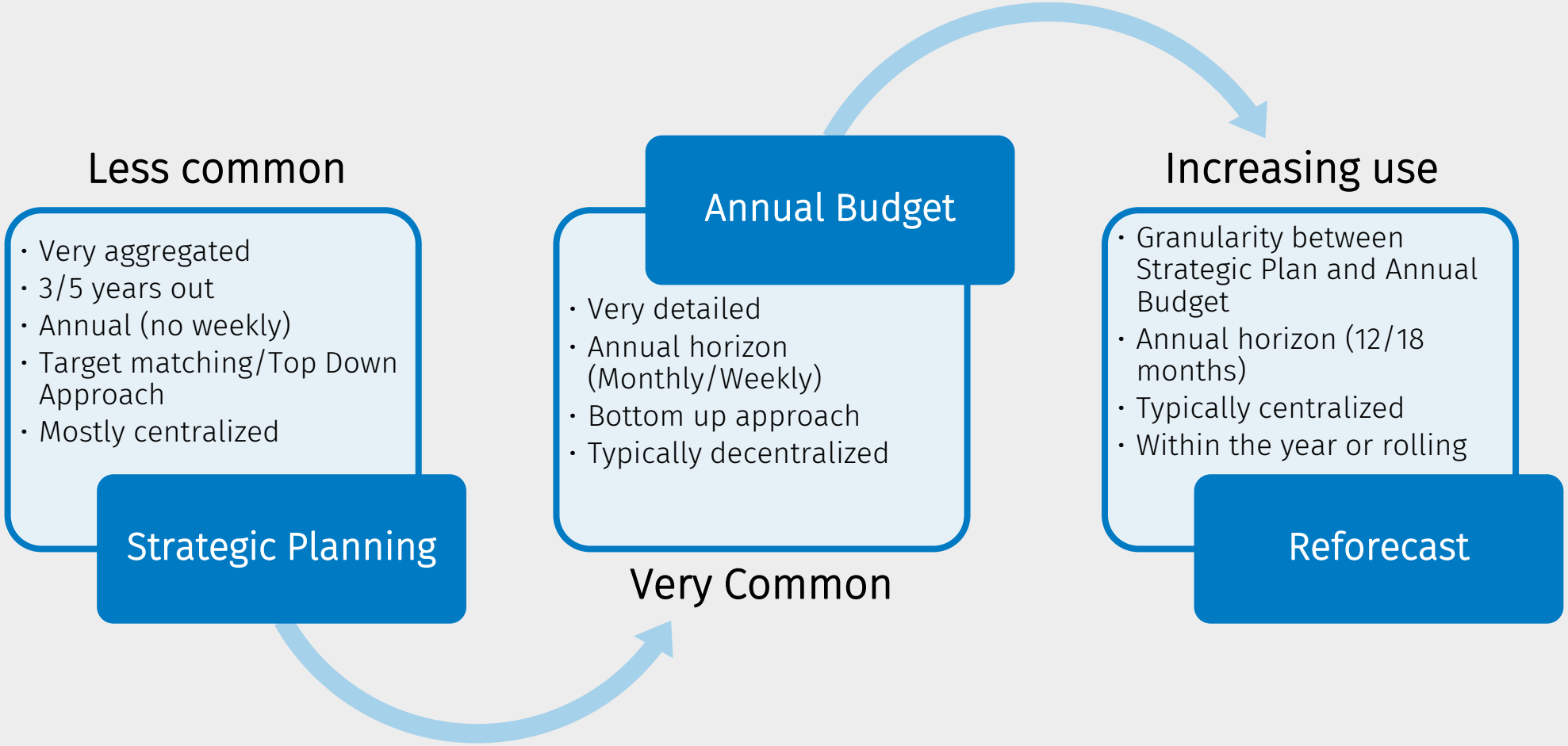
Funding

- It ensures funding for daily and strategic business operations
- It directs capital expenditure in the most profitable direction

Efficiency

- It increases production efficiency, controls costs and sets accountability
- It coordinates the activities of different departments in the organization

Planning & Forecasting – the how



Strategic vs operational plan

Strategic plan

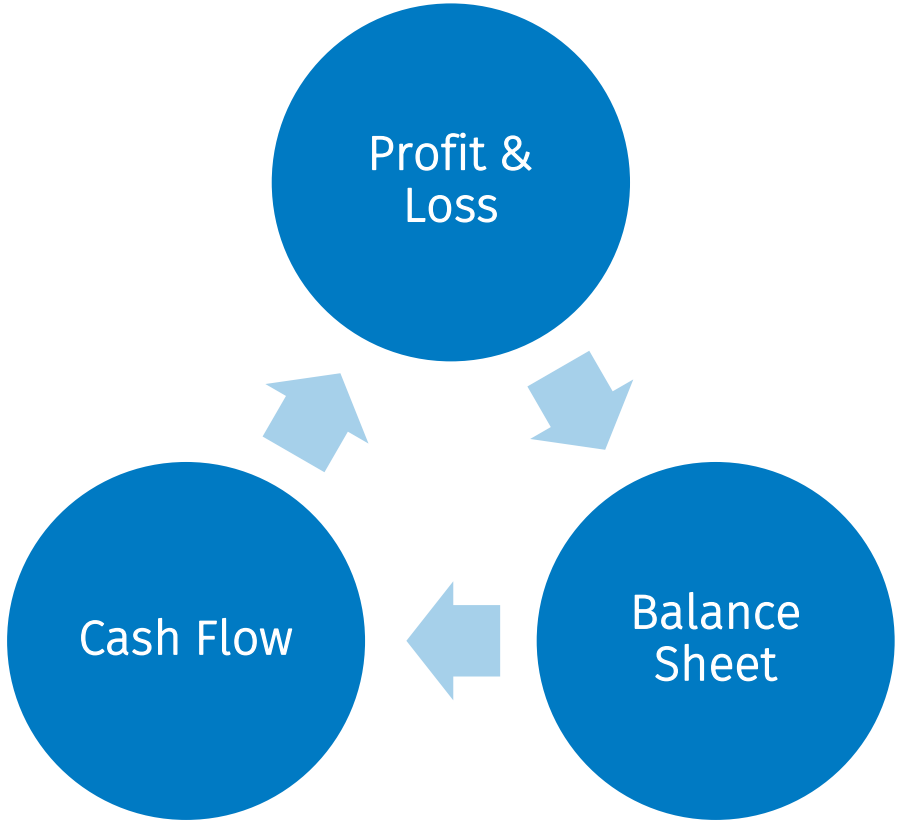
- Describes
  - high-level goals
  - long-term vision
  - organizational mission
- Not focused on implementation

Operational plan

- Smaller time horizon and scope
  - to provide a more granular view
  - to achieve strategic goals
- Focused on implementation in the form of actions and specific KPI, etc.

Planning and forecasting – Closing the Loop

Profit & Loss is the main but not the only planning object.



3 common budgeting methods

Incremental

- Adding or subtracting percentages to last year's actual amounts

Zero based

- Rebuilding budget figures from scratch each new year

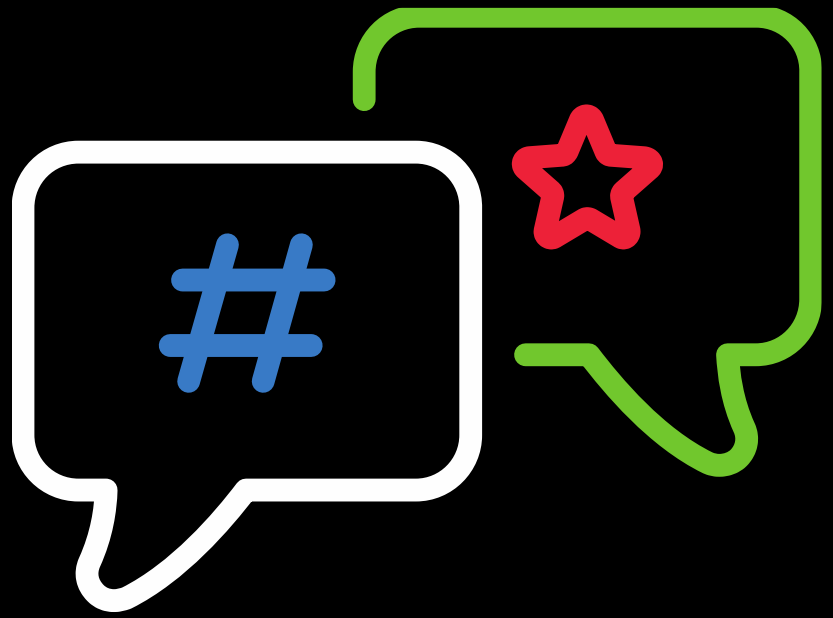
Activity based

- Linking activities with relative costs to reach target

Planning on CCH Tagetik

- Connect strategy, finance and operations to drive a profitable growth
- Align finance, sales, production, marketing, workforce, IT, etc.
- Analytic Information Hub allows you to manage different data models suitable for any planning process function
- Predictive intelligence provides estimates based on machine learning
- Financial workspace provides built-in tools for financial planning

Let’s talk about: Data model trends



Workspace

- Analytic Information Hub is the tool for planning
- Financial workspace is used for
  - Comparisons between scenarios at aggregated level
  - Budget consolidation
  - Cash flow planning
  - Simple models (planning at account level)

Datasets classification based on goals

Dataset structure can be tailored and optimized

Standard datasets

- Data entry
- Calculation
- Reporting

Reference datasets

- Configuration: definition of elements, rules, calculations...
- Transformation: look up tables...
- Assumptions: coefficients, data to be applied...

Datasources

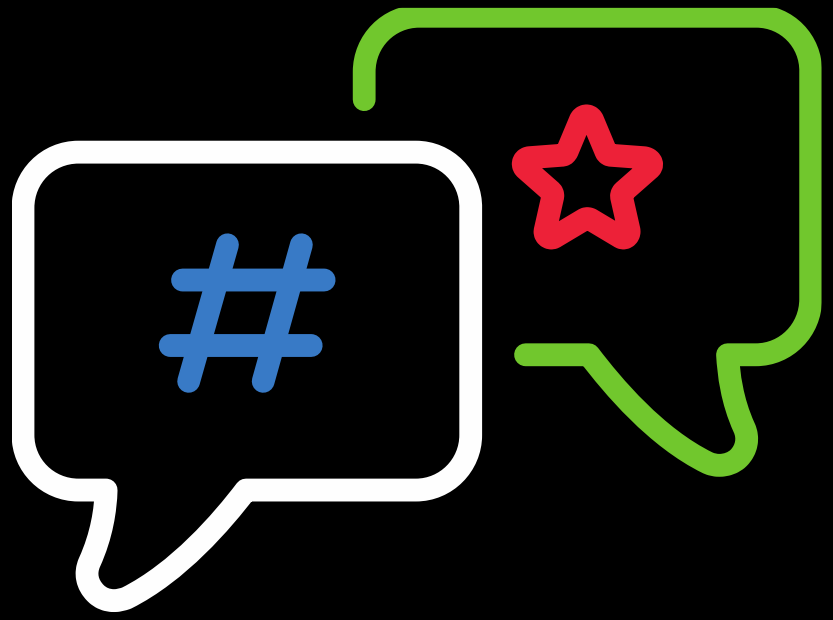
- Retrieve data from other workspaces
- Prepare data for
  - calculations
  - reports

Virtual datasets

Prepare data for

- Calculations
- Reports
- Financial Workspace

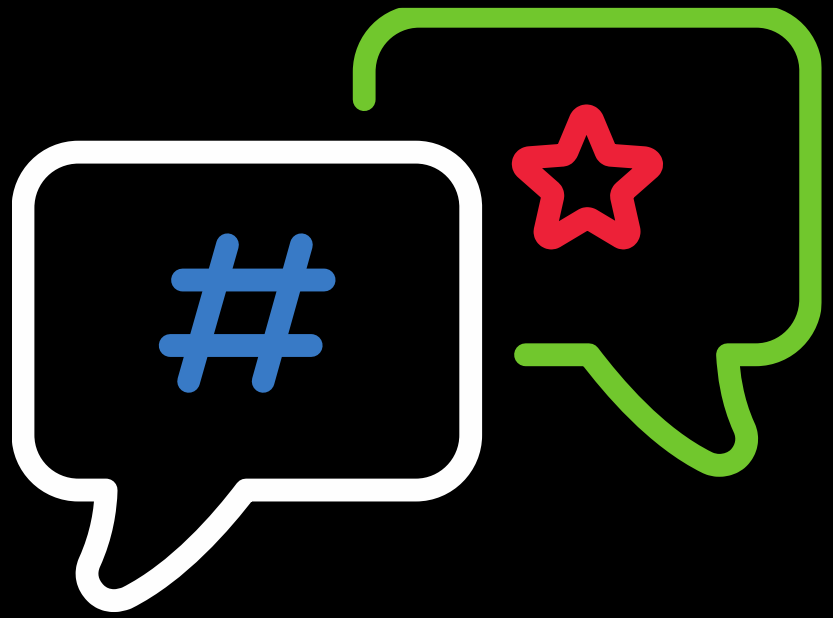
Let’s talk about: Business case



Business case

Financial view	Approach	Dimensions	Logic
1. Sales target	Top-down	Entity, Line of business	• Insert aggregated annual target • Distribute among products
2. Sales plan	Bottom-up	Salesrep, Product	• Insert annual quantity sold • Split on periods based on last actual • Adjust figures
3. Operating costs	Driver based	Account, cost center	• Use drivers • Apply markup • Manual adjustment
4. Personnel costs	Assumptions	Account, cost center, grade	• Plan headcount by position • Apply average global assumption to calculate financials
5. Capital expenditures	Financial impact of assets	Account, asset class	• Define amortization logics • Plan asset purchase • Assess the impact on financial statement

Let’s talk about: Top-down planning



Top-down planning: from general to specific

Company sets goals at aggregated level

Strategy is communicated and addressed in specific phases

Objectives are broken down to lower level through a given hierarchy

1. Sales target workflow

Company defines a sales target

- by Entity
- by Line of Business (aggregation of product)

Target amount must be broken down and assigned to

- Sales Rep
- Single Product

(based on last actual)

Assumption: currency not managed

Workshops Theory



DTP operations to:

- Split data across dimensions (allocations)

Other tools to distribute amounts

- Spreading
- Break back (recap)

Datasource to:

- Retrieve and prepare data to process

1. Sales target – Necessary ingredients

Data entry for target

- Standard dataset

Last actual to break down target

- Datasource to retrieve actual data from another workspace

Breakdown calculation

- DTP activity with allocation operation
- Standard dataset to store results

Dataset
10 Global target

Partitioning

- by business cycle

Fields

- Scenario, Period, Entity, Line of Business, Amount.
- Line of Business is not specified as hierarchy of product but as independent dimension

Dimension representing hierarchy nodes

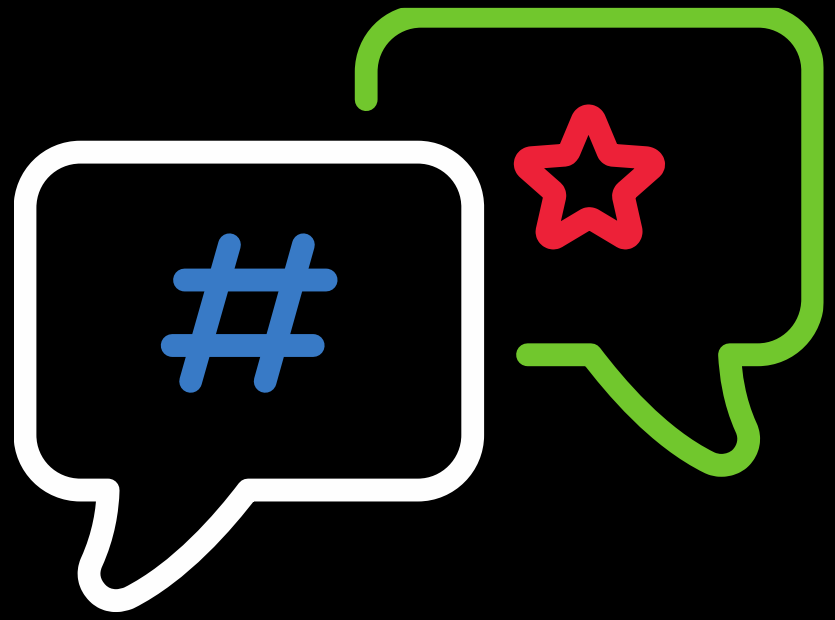
In case of:

- Dimension A with many elements
- Use of aggregation in:
  - DTP calculation (allocations)
  - Dataset (data entry or virtual dataset)

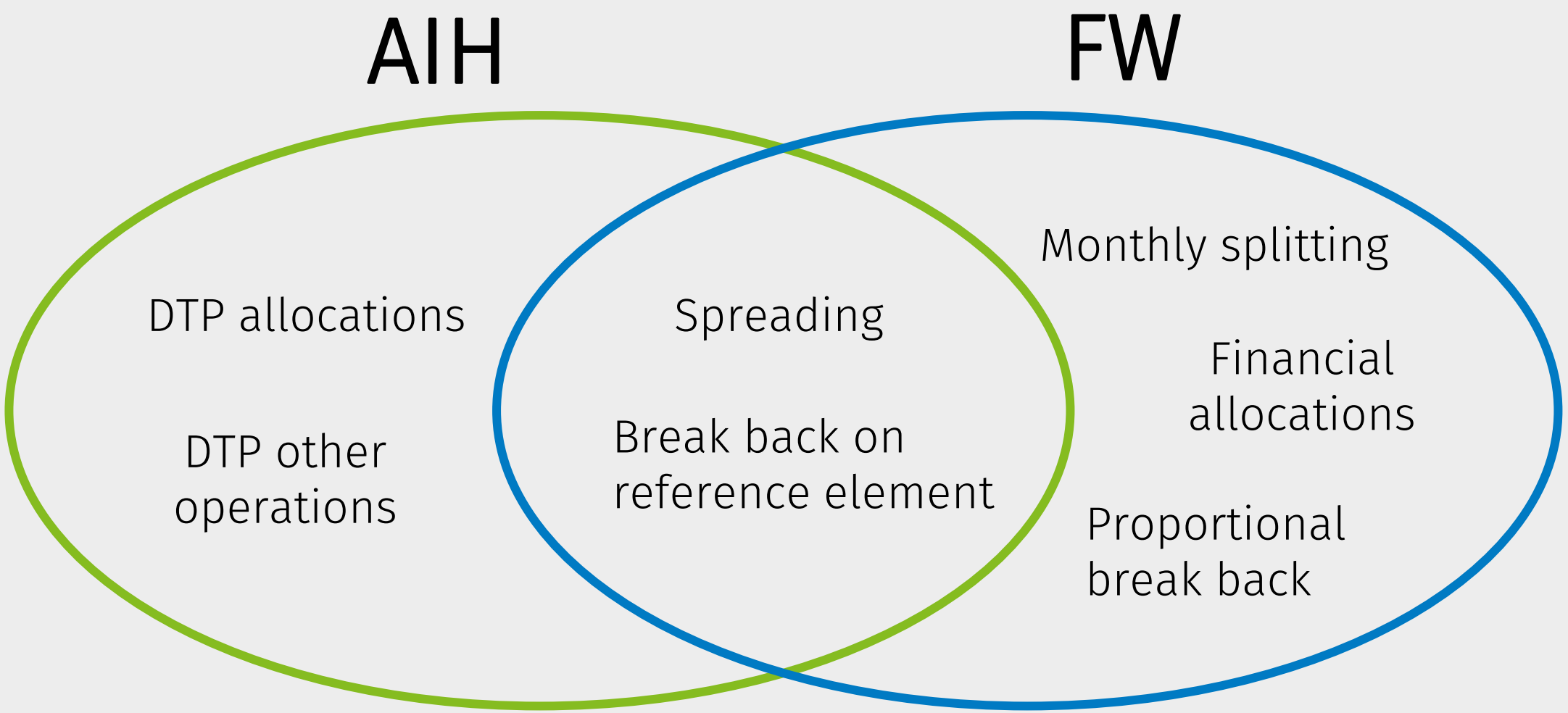
It is recommended

- to manage aggregation on A through the use of another dimension B
- The elements of the B represent the hierarchy nodes of A (coding should be coherent)
- Association across dimensions is done through datasource reading from product hierarchy

Let’s talk about: Amount distribution



Tools to distribute amounts



Spreading in data entry form

Allows you to change result of formula and “spread” the difference on all elements of the formula.

	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter	Total
A01 - Antivirus	10.00	20.00	50.00	20.00	100.00



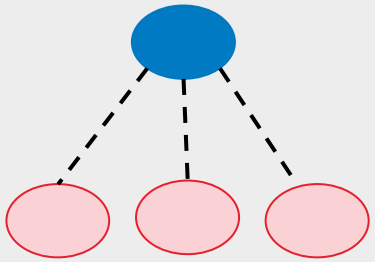
Spreading methods

- Proportional: existing amounts used as weight
- Flat: distributed evenly
- Seasonality pattern: 4-4-5, 4-5-4, 5-4-4, Custom
- Based on a different scenario

Spreading

When to use it?	When adjustment is not done on months but on year, is not done on elements, but on total
Who?	End User in Data Entry mode. No audit trial
What is it ?	Feature allows user to change value in a formula and difference input is spread over operands
Spreading methods	Flat, Proportional, Reference scenario, Season pattern

Break-Back on AIH (recap)

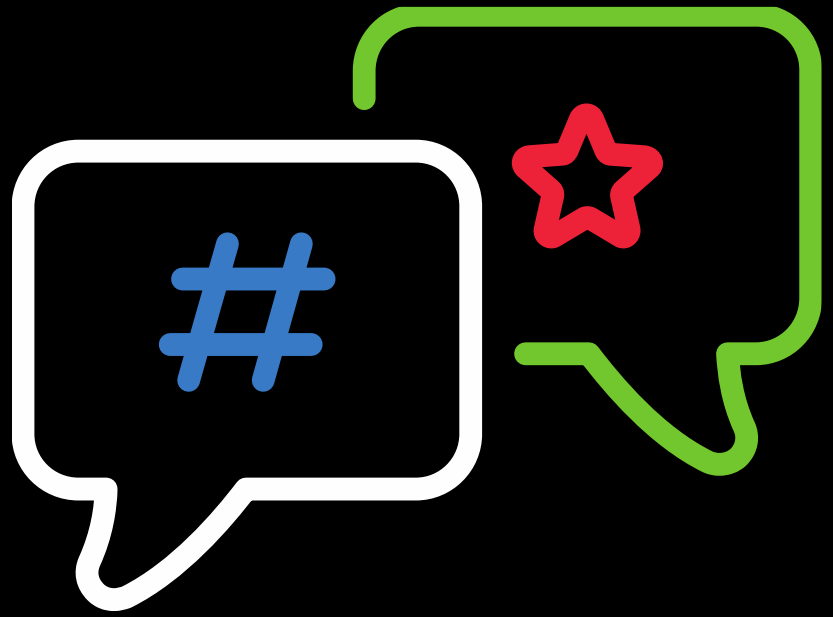


- Used when Top Down approach is necessary to draft Budget
- Allows you to enter data at hierarchy node level in Data Entry forms
- Amount will be saved on reference elements
- Available in multidimensional matrices

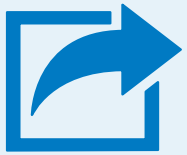
Spreading vs Break back on AIH

	Spreading	Break back
Setting management	No setting	General/Process/Matrix
Source data	Formulas related to editable cells	Hierarchy nodes of dimension enabled
Target data	Formula addends (elements)	Reference element of the node
Pros	Flexible and ‘out of the box’	Saving at aggregated level
Cons	No audit	Not a real distribution tool

Let’s talk about: Allocations

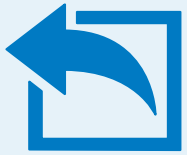


Allocation Basic Terms



Sender

- Starting data not distributed
- Dataset with data to be allocated



Driver

- Coefficients to distribute amounts
- Driver dataset



Receiver

- Target with amounts distributed among the required dimensions
- Dataset with allocated data

Why should allocations be used?

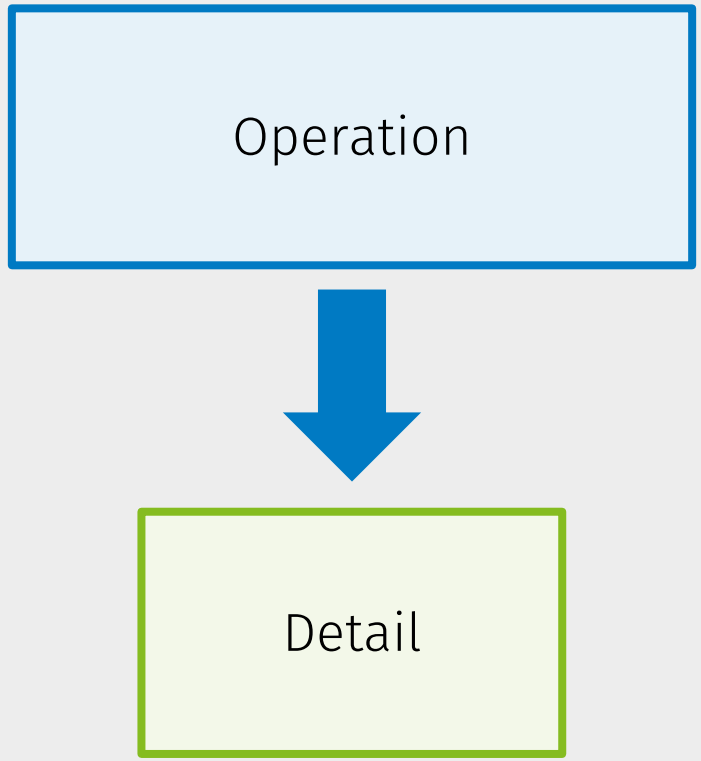


- To distribute a top-down adjustment value to other dimensions
- To allocate a value stored by generic elements to specific elements
- To charge generic costs and assess product profitability
- To assign resources (financial, assets, human) to support organization goals

Allocation operation structure

- OPERATION
  - Driver dataset
  - Dataset data to be allocated (sender)

- DETAIL
  - Measure to allocate
  - Driver measure
  - Dimensions on which to allocate



Allocation use restrictions



Matching field not present – coherent dimensions and coding are recommended

- Receiver amount must have the same code as sender amount
- Receiver dimensions are copied from driver dataset fields with same dimensional type
- All other dimensions are copied from sender fields with same dimensional type
- All other measures are copied from sender fields with the same code

Not suitable for allocating more measures in a single operation

Not suitable for allocating and reading driver across Scenario, Period, Entity

- If necessary to allocate on them -> manage on another dimensions
- If necessary to use drivers from other scenario-period-entity -> manipulate and match

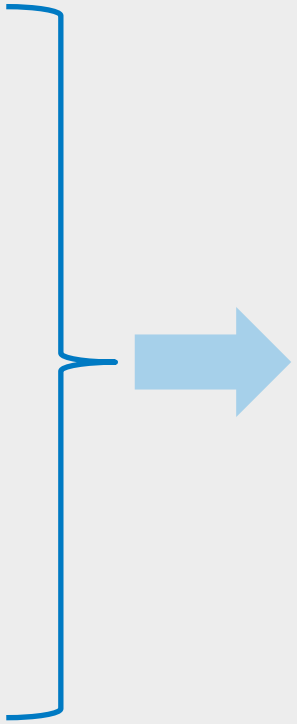
Allocation: simple example

Data to allocate

Account	Cost Center	Amount
Energy	NA	1000

Driver

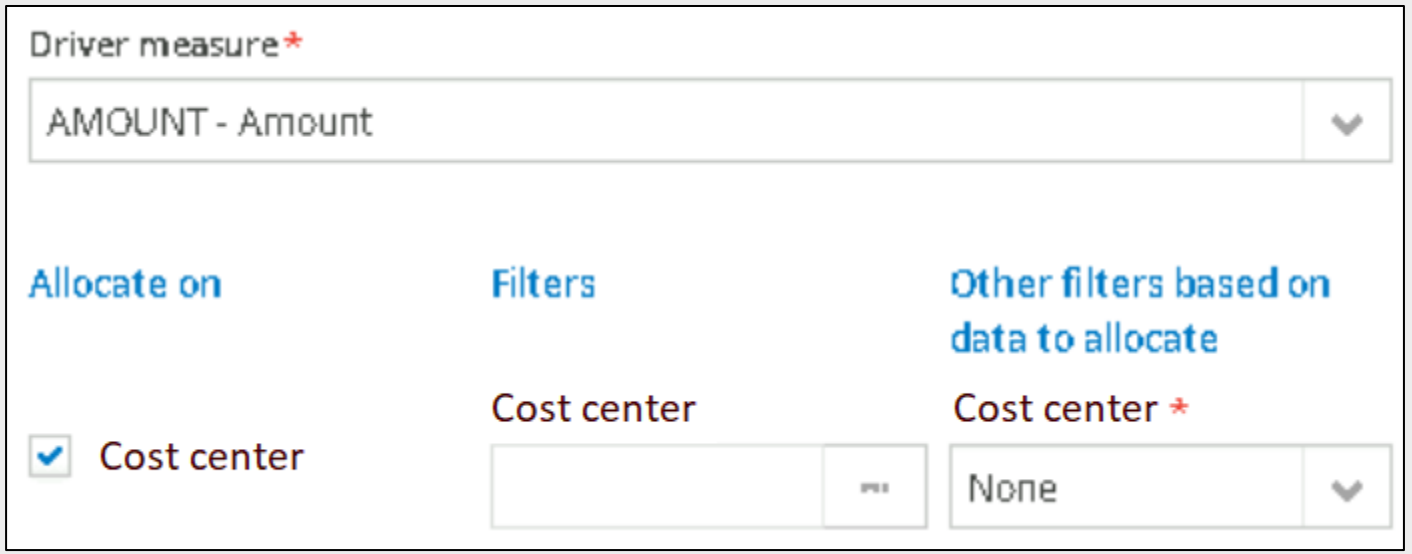
Account	Cost Center	Amount
Headcount	Administration	10
Headcount	HR	20
Headcount	Marketing	35
Headcount	Sales	35



Rows generated by Allocation

Account	Cost center	Amount
Energy	Administration	100
Energy	HR	200
Energy	Marketing	350
Energy	Sales	350

Simple Example – DTP Allocation detail configuration



Used to decide how to reduce the subset of elements on which the allocation will occur

Allocation: a more complex case

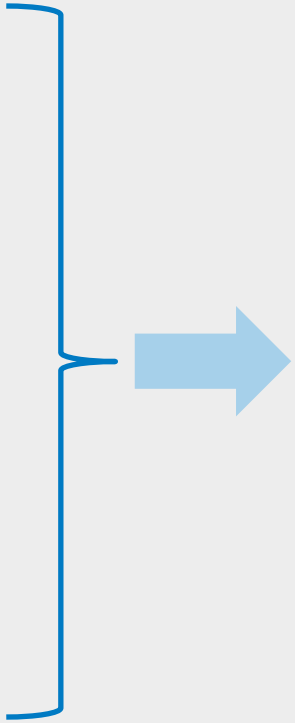
Allocation must occur within each line of business

Data to allocate

LoB	Amount
Hardware	600
Software	400

Driver

Sales	LoB	Product	Amount
S001	Hardware	Keyboard	10
S001	Hardware	Screen	20
S001	Software	Oper.Sys.	35
S001	Software	Writer	35



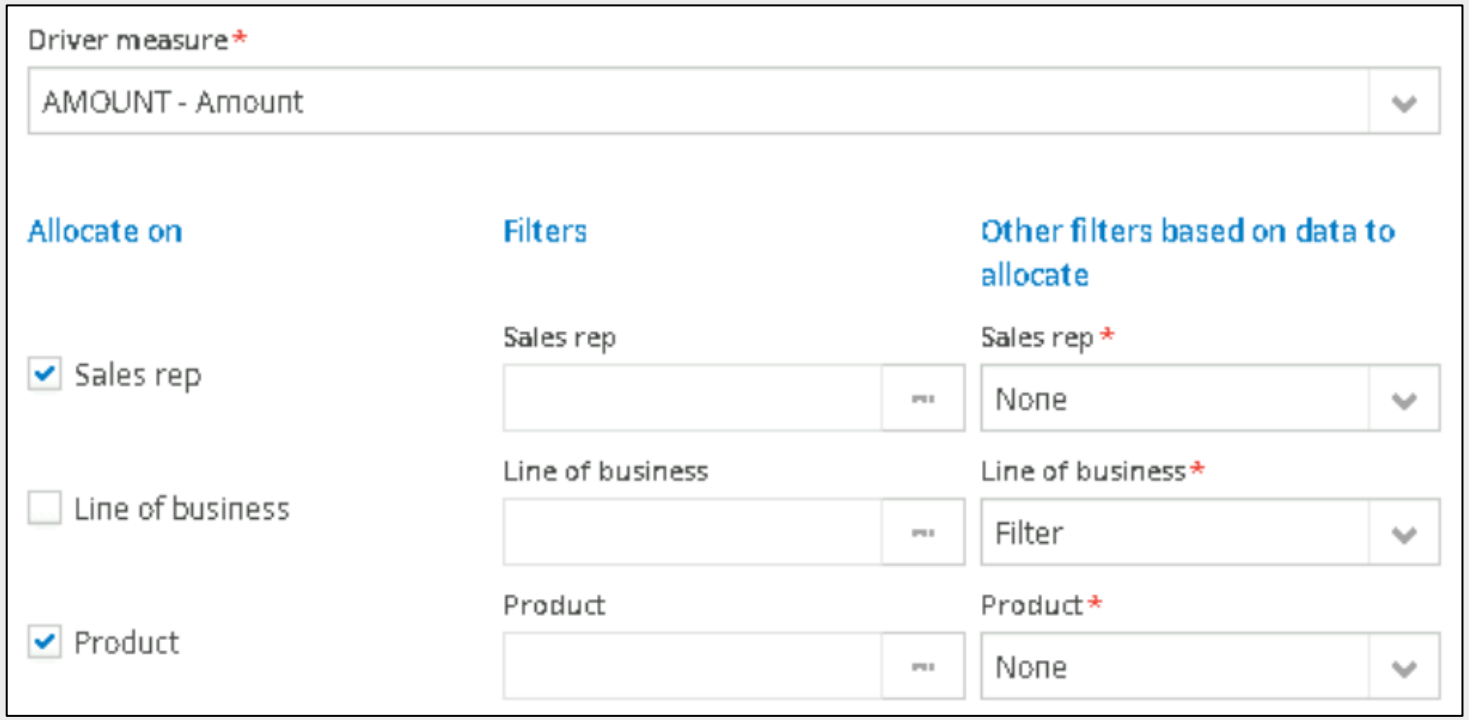
Rows generated by Allocation

Account	LoB	Product	Amount
S001	Hardware	Keyboard	600*10/30= 200
S001	Hardware	Screen	600*20/30= 400
S001	Software	Oper.Sys.	400*35/70= 200
S001	Software	Writer	400*35/70= 200

Denominator of driver coefficient is considered for each line of business

A more complex case – DTP Allocation detail configuration

How can we achieve the result in the example?



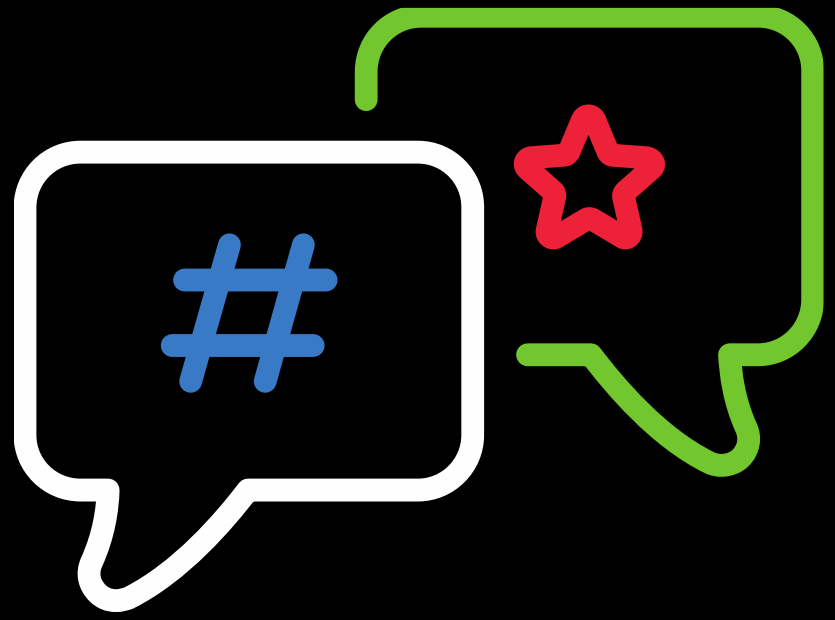
Used to decide whether to group or not. Allocation must occur within these elements

Now it’s your turn

Exercise # 1



Let’s talk about: Bottom-up planning



Bottom-up planning: from specific to general

Plan is prepared at the lowest levels of the organization in a detailed way

Goals are developed, aggregated and funneled through consecutive levels until top management

2. Sales planning workflow

- Sales Reps input their budgeted annual sales which should at least meet the target they have received
- Annual amount must be split on
  - months

  based on last actual
- Monthly figures can be further adjusted manually in a data entry structure with 12 columns
- Figures will be available in reporting, aggregated and approved at different levels

Workshops Theory



DTP operations to:

- Split data across time (pivoting)

Order of DTP operations -> efficiency

Datasource to:

- Retrieve and prepare data to process

2. Sales planning – necessary ingredients

Data entry for annual planning

- Standard dataset

Last actual for monthly splitting %

- Datasource to retrieve actual data from another workspace

Monthly splitting calculation

- DTP activity with a sequence of enrichment operations
- Standard dataset to store results

Dataset
21 Monthly sales planning

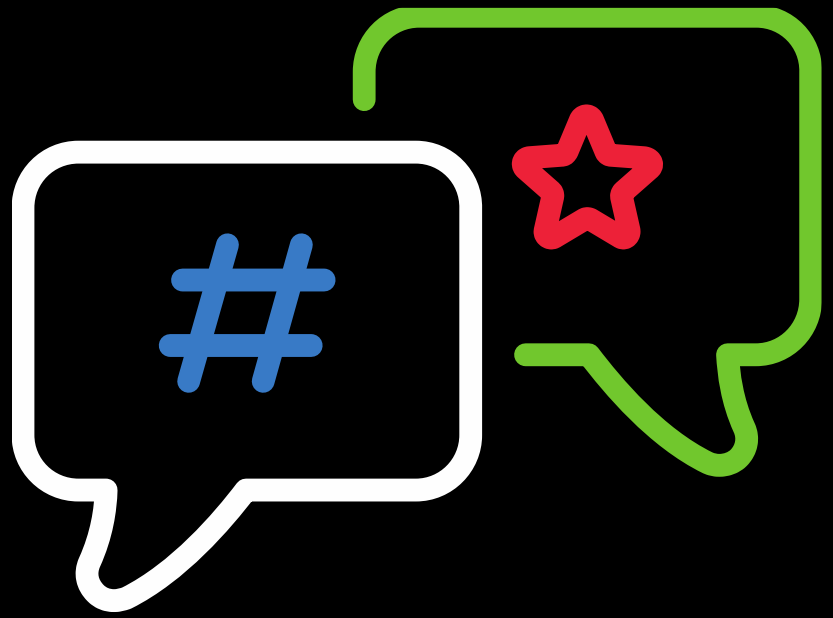
Partitioning

- by business cycle

Fields

- Scenario, Entity, Sales Rep, Product, Total
- P01 to P12 to store the amounts of each month
- Total represents the annual total, necessary for calculations
- In case of required multidimensional reporting, period dimension is present and set to 12
- Monthly amounts on columns is convenient for:
  - Data entry (less rows)
  - Making easier calculation

Let’s talk about: Monthly splitting with DTP operations



Monthly splitting DTP sources

What should be inserted first?

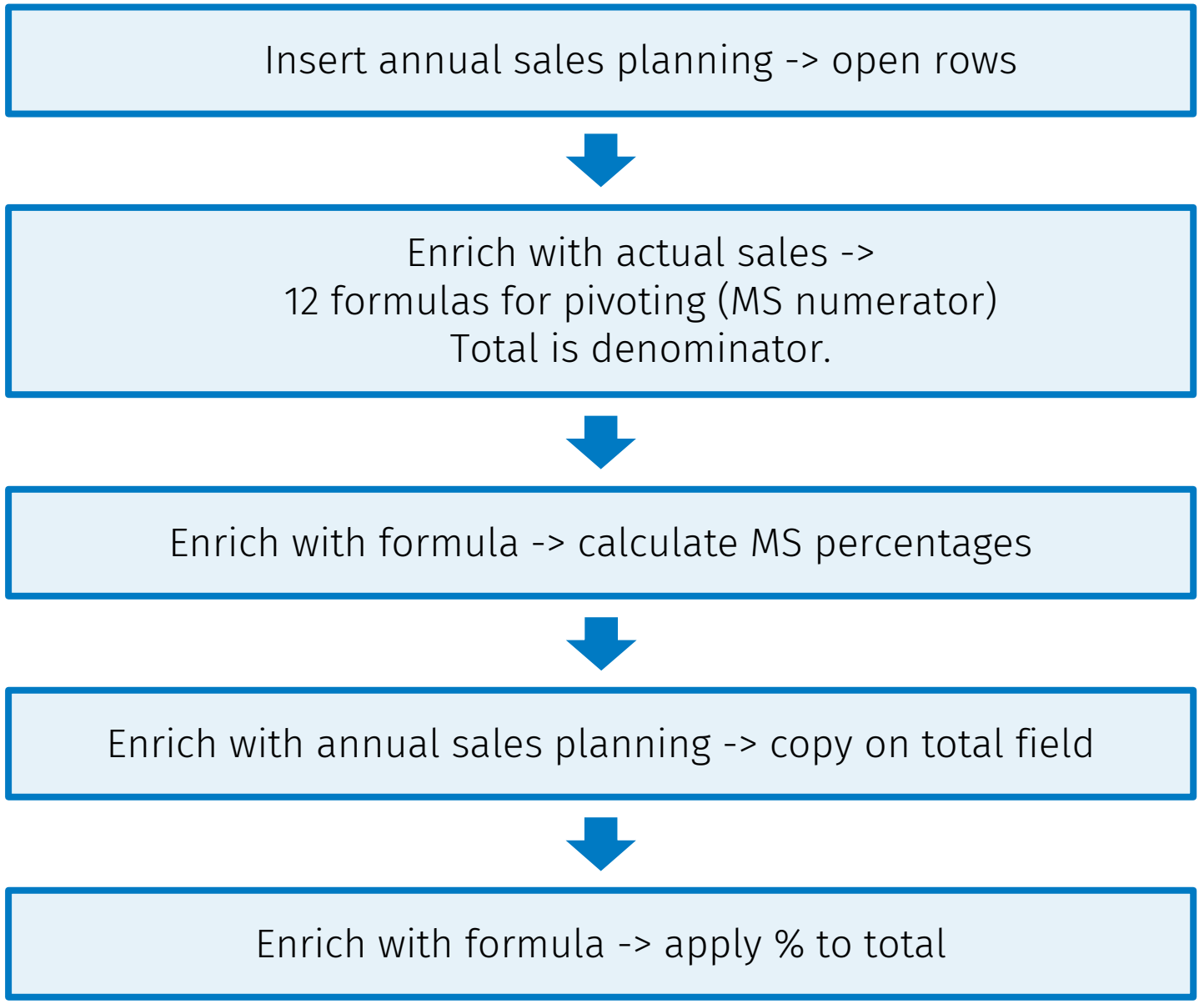
Datasource: last actual sales

Dataset: annual sales planning

- The order is relevant both logically and for efficiency
- Ask yourself what the minimum necessary set of rows is

- All the rows from annual sales planning are required even though calculation starts from last actual sales.
- We first open the rows with the dimensions of annual sales planning.

Monthly splitting DTP operations



Operation 1 – Insert annual sales planning

Actual sales

Product	Period	Amount
Keyboard	1	30000
Keyboard	2	10000
...	...	...

Annual sales planning

Product	Amount
Keyboard	180000
Screen	60000
Writer	230000

Target

Product	P01	P02	P03	Total

Operation 1 – Insert annual sales planning

Actual sales

Product	Period	Amount
Keyboard	1	30000
Keyboard	2	10000
...	...	...

Annual sales planning

Product	Amount
Keyboard	180000
Screen	60000
Writer	230000

Target

Product	P01	P02	P03	Total
Keyboard				
Screen				
Writer				

Operation 2 – Enrich with last actual sales

Actual sales

Product	Period	Amount
Keyboard	1	30000
Keyboard	2	10000
…	…	…

Annual sales planning

Product	Amount
Keyboard	240000
Screen	60000
Writer	250000

- Join on product (condition)
- Copy period in 12 columns using proper filters (attribute)
- Copy the sum of amounts in total, grouping by product

Target

Product	P01	P02	P03	Total
Keyboard	30000	10000	20000	180000
Screen	15000	5000	10000	50000
Writer	8000	5000	40000	200000

Operation 3 – Enrich with formula for monthly splitting %

Actual sales

Product	Period	Amount
Keyboard	1	30000
Keyboard	2	10000
...	...	...

Annual sales planning

Product	Amount
Keyboard	240000
Screen	60000
Writer	250000

- Divide each month by the total

Target

Product	P01	P02	P03	Total
Keyboard	16.67%	5.56%	11.11%	180000
Screen	30%	10%	20%	50000
Writer	4%	2.5%	20%	200000

Operation 4 – Enrich with annual sales planning

Actual sales

Product	Period	Amount
Keyboard	1	30000
Keyboard	2	10000
…	…	…

Annual sales planning

Product	Amount
Keyboard	240000
Screen	60000
Writer	250000

- Join on product (condition)
- Copy total (attribute)

Target

Product	P01	P02	P03	Total
Keyboard	16.67%	5.56%	11.11%	240000
Screen	30%	10%	20%	60000
Writer	4%	2.5%	20%	250000

Operation 5 – Enrich with formula apply % to total

Actual sales

Product	Period	Amount
Keyboard	1	30000
Keyboard	2	10000
...	...	...

Annual sales planning

Product	Amount
Keyboard	240000
Screen	60000
Writer	250000

- Multiply each month for the total

Target

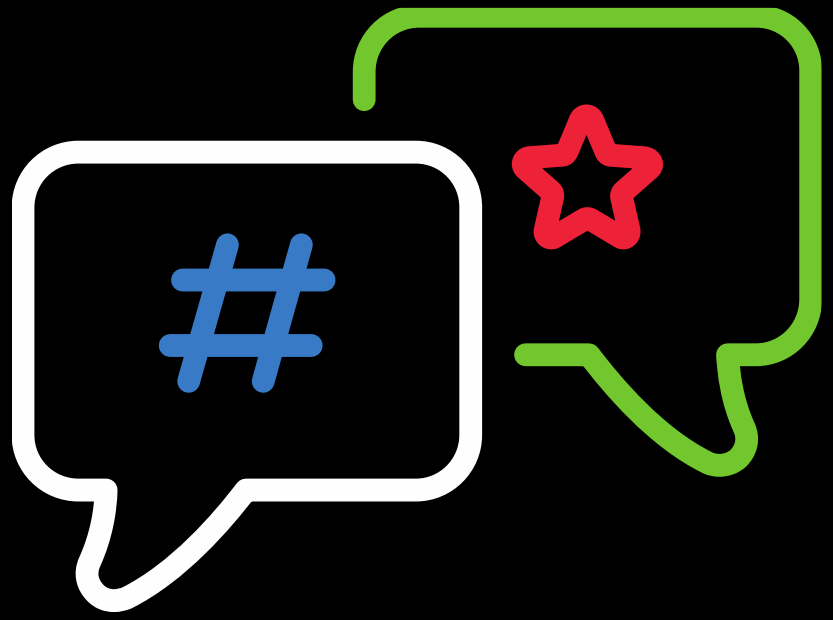
Product	P01	P02	P03	Total
Keyboard	40000	13333	26666	240000
Screen	18000	6000	12000	60000
Writer	10000	6250	50000	250000

Now it’s your turn

Exercise # 2



Let’s talk about: Driver based planning



Driver based planning

Driver-based planning focuses on identifying the key factors that drive the business and create a plan based on those drivers.

It is necessary to understand how the key drivers lead to actionable outcomes and results.

Finding Drivers is a challenge

In Driver based planning the most important question is:

What drives Revenues or Costs ?

Type of business and type of costs

Big research of data can provide answers
Use of machine learning could be helpful

3. Driver based planning workflow

Budget baseline is actual of previous year that can be modified through

- A markup
- The effect of different drivers
- A manual overwrite

Drivers can be applied to

- Accounts
- Cost centers

or the relative aggregations

Assumption: period not managed

Workshops Theory



Reference datasets for

- Configuration of synthetic definition with priority and exceptions

  (Level)

DTP operations to:

- Transform the baseline according to configurations

Datasource to:

- Retrieve and prepare data to process

3. Driver based planning – necessary ingredients

Baseline

- Datasource to retrieve actual data from financial workspace

Data entry

- Standard dataset for drivers
- Reference dataset for cost planning definitions (markup, drivers, overwrite)

Cost planning calculation

- DTP activity with a sequence of enrichment operations to apply markup, drivers and overwrite
- Standard dataset to store results

Dataset
32 Cost planning definition

Dataset type

- reference dataset

Partitioning

- by business cycle

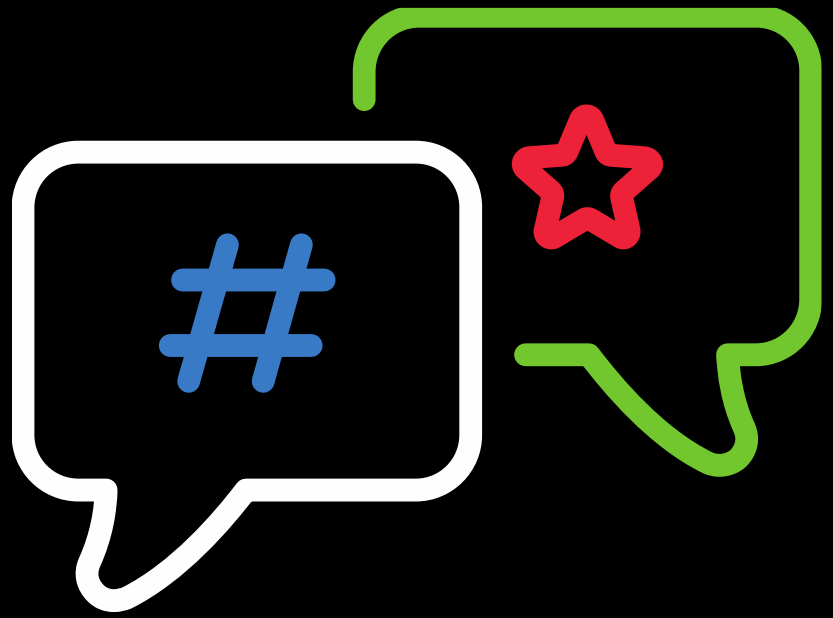
Fields

- Scenario, Period, Entity, Account, Cost center, Markup, Overwrite
- The sequence represent the order of application of the rules in case of conflicts
- It is possible to specify 2 drivers, recalled through a dictionary. The logic does not change with more drivers

Using wild card or leaving fields empty

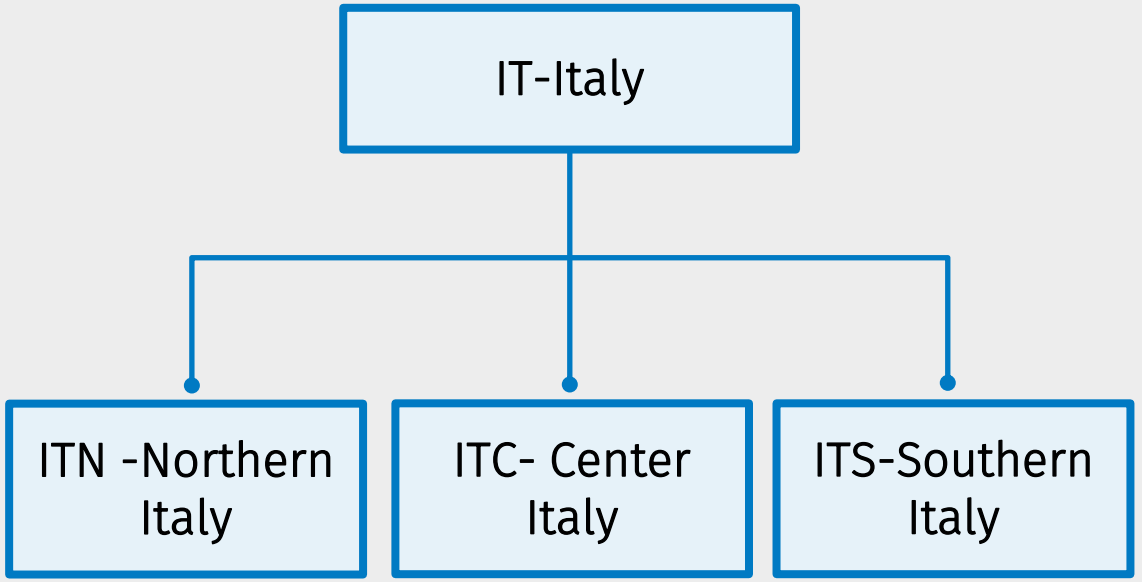
- it is possible to have a synthetic definition (e.g. for all cost centers, for a group of account, …)

Let’s talk about: Reference type level in reference dataset



Levels in reference dataset

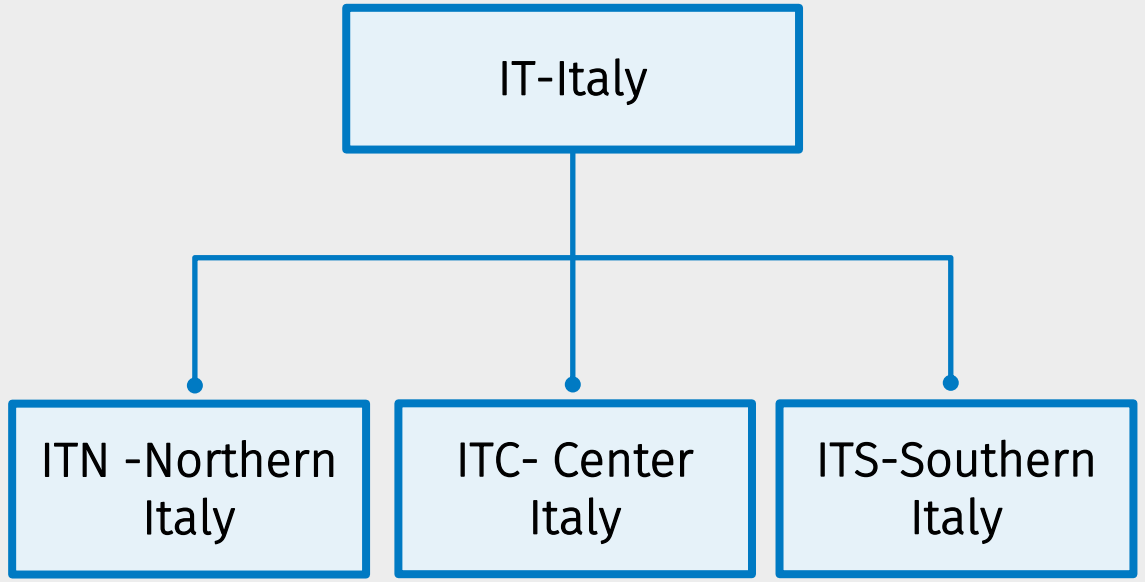
Region (Condition)	Level	Price (Attribute)
IT%	1	1000
ITN	2	1300



Scenario	Period	Entity	Region	Price
2050FCT	06	B00	ITN	
2050FCT	12	B00	ITS	
2050FCT	10	B00	ITC	

Levels in reference dataset

Region (Condition)	Level	Price (Attribute)
IT%	1	1000
ITN	2	1300

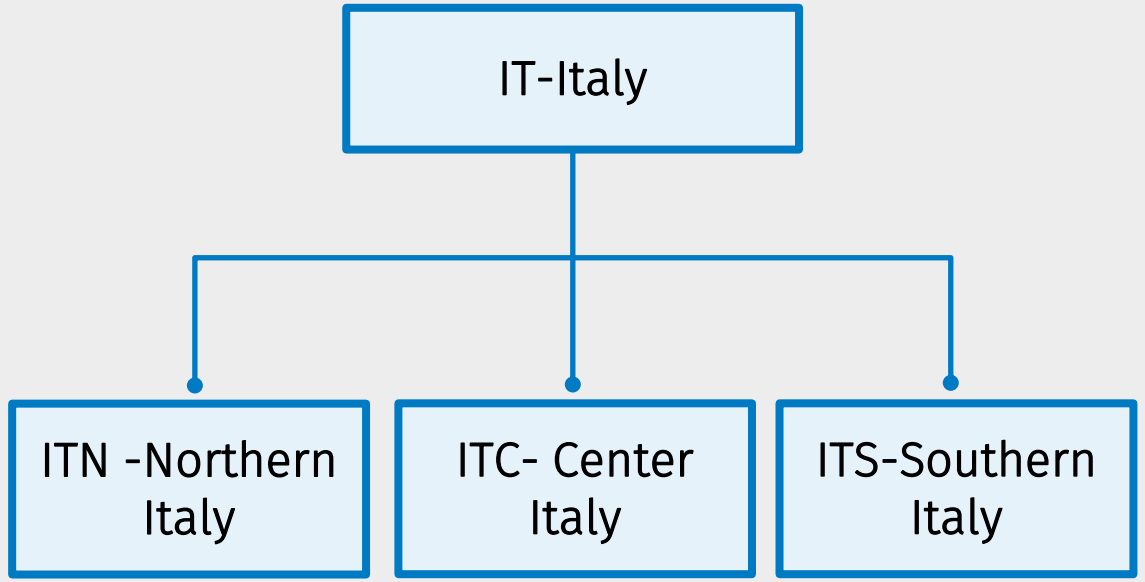


Scenario	Period	Entity	Region	Price
2050FCT	06	B00	ITN	1000
2050FCT	12	B00	ITS	1000
2050FCT	10	B00	ITC	1000

Level 1 affects all Regions whose code starts with IT

Levels in reference dataset

Region (Condition)	Level	Price (Attribute)
IT%	1	1000
ITN	2	1300



Scenario	Period	Entity	Region	Price
2050FCT	06	B00	ITN	1300
2050FCT	12	B00	ITS	1000
2050FCT	10	B00	ITC	1000

Level 2 affects rows whose code is ITN

Out of dictionary value allowed

By default, if a text field has a dictionary associated, it is possible to select only elements from the dictionary

For dimensions it is only possible to select dimensional elements

How to use a drop-down list and be able to specify exception, e.g. select all accounts starting with code 50…?

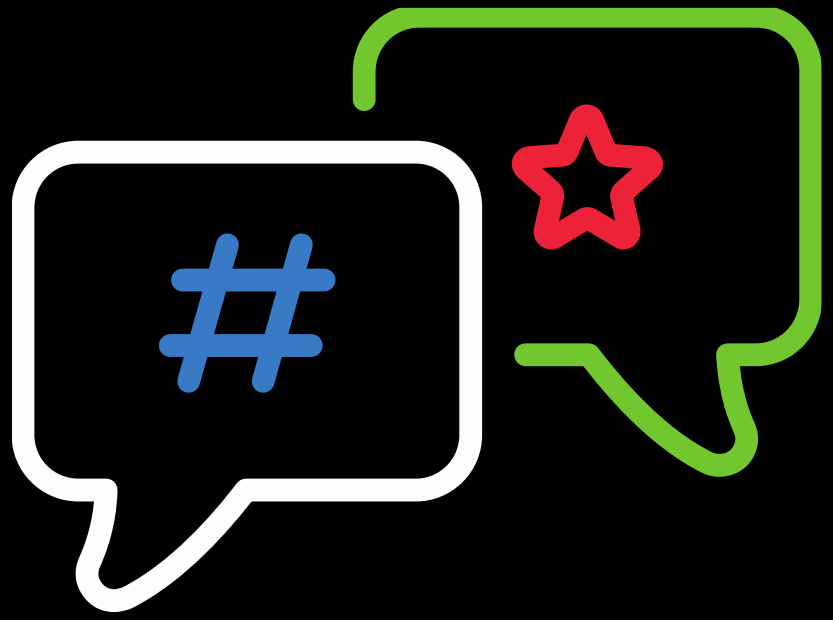
- Specify the field relative to account dimension as text field
- Associate a suitable dictionary
- Flag the “out of dictionary values allowed”
- Drop-down list is available, but it is possible to overwrite with any value

Now it’s your turn

Exercise # 3



Let’s talk about: Driver based calculation with DTP operations



Cost planning DTP sources

What should be inserted first?

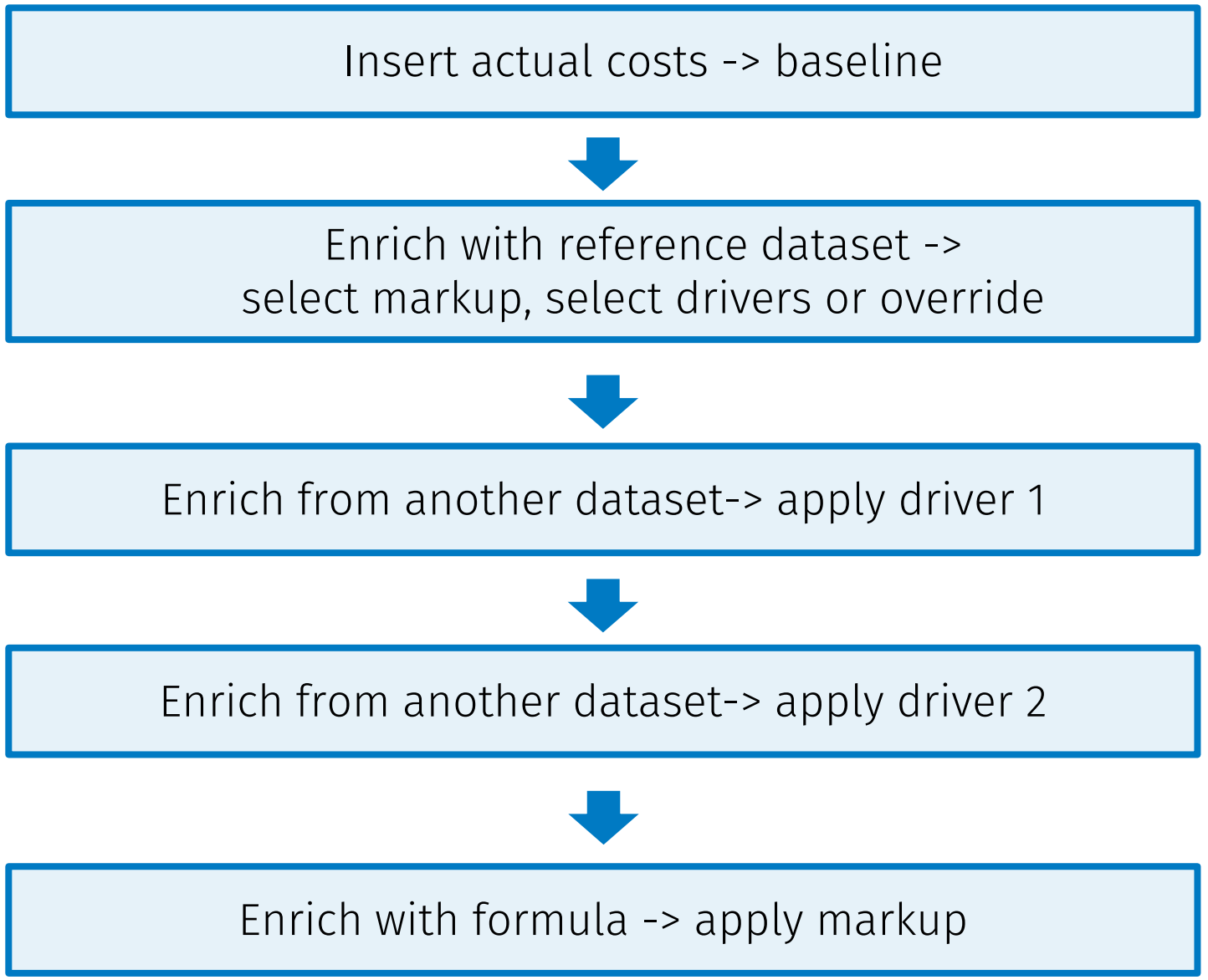
Datasource: last actual costs

Dataset: Driver definition

Reference dataset: cost planning definition

- Actual costs
- All the rows from actual costs are required as they are the baseline.

Cost planning DTP operations



Operation 1 – Insert actual costs

Cost planning definition

Account	MUP	Driver	Overwrite
Energy		Manufacturing cost	
Rent			10000
Insurance	10%		…

Actual costs

Account	Amount
Energy	20000
Rent	8000
Insurance	15000

Target

Account	Amount PY	Driver	%	Amount
Energy	20000			
Rent	8000			
Insurance	15000			

Operation 2 – Enrich with cost planning definition

Cost planning definition

Account	MUP	Driver	Overwrite
Energy		Manufacturing cost	
Rent			10000
Insurance	10%		…

Actual costs

Account	Amount
Energy	20000
Rent	8000
Insurance	15000

- Join on account (condition)
- Copy definitions (attributes)

Target

Account	Amount PY	Driver	%	Amount
Energy	20000	Manufacturing cost		
Rent	8000			10000
Insurance	15000		10%	

Operation 3 – Enrich with driver definition

Cost planning definition

Account	MUP	Driver	Overwrite
Energy		Manufacturing cost	
Rent			10000
Insurance	10%		…

Driver defintion

Driver	%
Manufacturing cost	5%
Sales effect	8%
…	…

- Join on driver (condition)
- Copy % (attributes)

Target

Account	Amount PY	Driver	%	Amount
Energy	20000	Manufacturing cost	5%	
Rent	8000			10000
Insurance	15000		10%	

Operation 4 – Enrich with formula – apply %

Cost planning definition

Account	MUP	Driver	Overwrite
Energy		Manufacturing cost	
Rent			10000
Insurance	10%		…

Driver defintion

Driver	%
Manufacturing cost	5%
Sales effect	8%
…	…

- Apply % to Amount PY

Target

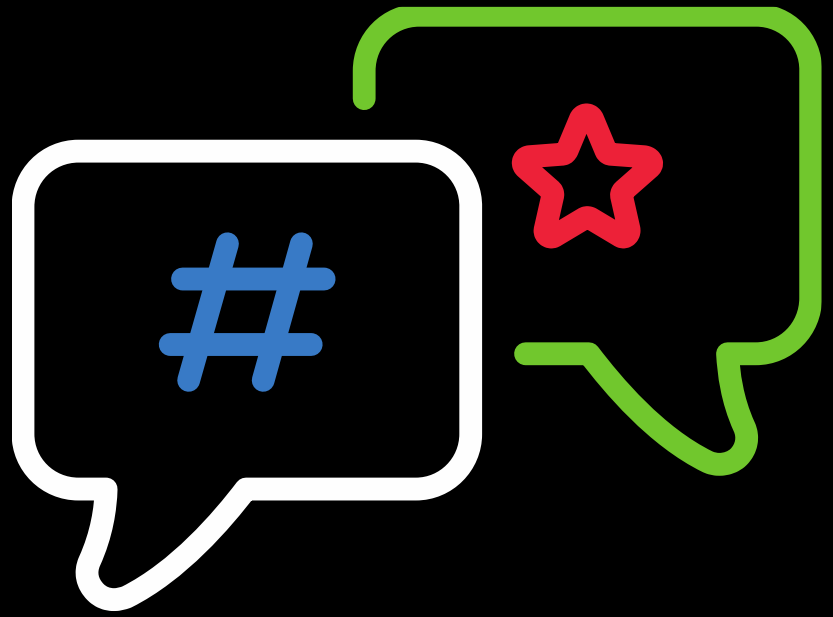
Account	Amount PY	Driver	%	Amount
Energy	20000	Manufacturing cost	5%	21000
Rent	8000			10000
Insurance	15000		10%	16500

Now it’s your turn

Exercise # 4



Let’s talk about: Assumptions



Assumptions

- At its simplest, a budget creates projections by adding assumptions to current data
- Companies use assumptions to enable quick changes in the budget and to support multiple planning scenarios.
- The organization can improve its expense control, budget accuracy and hiring plans and reduce manual input

- Assumption example in personnel cost planning:
  - number of hours worked
  - average hourly wage
  - average salary by position
  - salary % change
  - tax rates....

4. Personnel planning workflow

- Annual assumption on average amounts by grade are input
- Assumptions are applied to budgeted FTE by month (12 columns) to calculate the total monthly amounts of
  - Wages and salary
  - Contribution
  - Bonus
  - Benefits
- Figures will be unpivot and available in reporting

Workshops Theory



Reference datasets to

- Define assumption
- Define calculation rules
- Verticalize

Virtual dataset to:

- Retrieve, join data, assumption and calculation rules and prepare data to process

DTP operations to:

- Apply assumptions (enrichment with formula field)
- Unpivot data
- Group and delete

4. Personnel planning – necessary ingredients

Annual assumption

- Reference dataset with average amounts by grade/account

Headcount by position

- Standard dataset with FTE by CC/grade and month

Calculation definition

- Reference dataset for definition
- Virtual dataset for data preparation

Calculation execution

- DTP with 2 activities to verticalize and group
- Standard dataset to store the result

Dataset
40 Personnel assumption

Dataset type

- reference dataset - deploy

Partitioning

- by business cycle

Fields

- Scenario, Period, Entity, Grade
- 4 columns to store the average amounts of the 4 accounts

Account is omitted

- as there is no requirement for flexible account selection

Dataset
41 Headcount by position

Partitioning

- by business cycle

Used also as calculation target

Fields

- Scenario, Entity, Account, Cost center, Grade
- P01 to P12 to store monthly amounts
- Period and account to be used after unpivoting
- 4 columns to copy assumptions
- Formula to write calculation rules
- Amount to store unpivoted results

Dataset
42 Personnel calculation rules

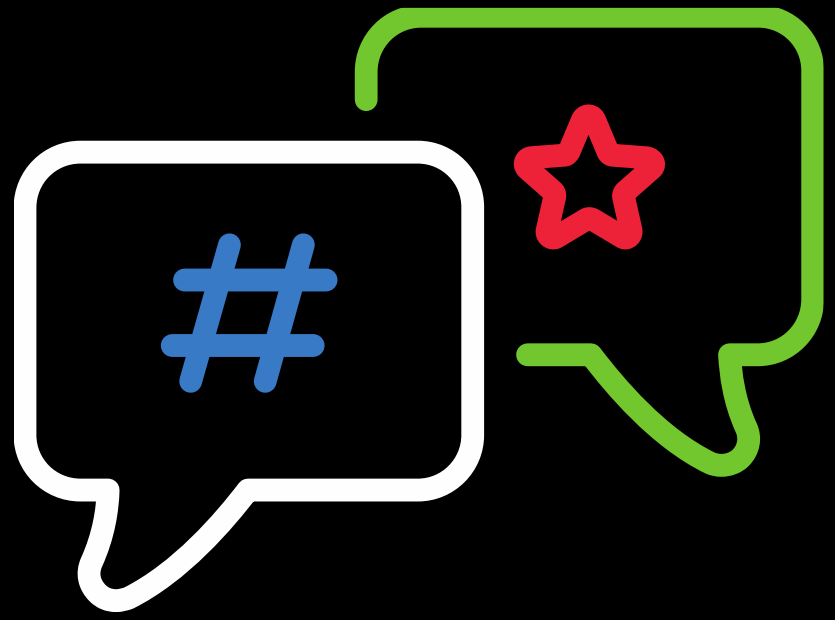
Partitioning

- by business cycle or none

Fields

- Period in, set arbitrarily to 12, used to join with other datasets to open rows
- Period, used for unpivoting and for calculations
- Account, used to define calculations

Let’s talk about: Calculation rules definition and application



Calculation input

For each period we want to calculate by country:

- Total wage = N. of FTE * daily wage * N. of working day
- Total contribution = Total wage * Contribution %

Country	FTE P10	FTE P11	FTE P12	Daily wage	Contributon %
France	70	69	75	100	20%
USA	90	80	80	150	10%
Japan	100	80	90	120	15%

Defining calculation rules

Result account = Factor 1 * Working day * Factor 2 * Coefficient

Country	Result account	Factor 1	Working days	Factor 2	Coefficient
France	Wage	FTE P10	22	Daily wage	1
France	Contribution	FTE P10	22	Daily wage	Contribution %
France	Wage	FTE P11	20	Daily wage	1
France	Contribution	FTE P11	20	Daily wage	Contribution %
France	Wage	FTE P12	18	Daily wage	2
France	Contribution	FTE P12	18	Daily wage	2 * Contribution %
…	…	…	…	…	…

Virtual dataset

- join input data and rules
- write a formula with the concatenation of the fields involved in calculation rules

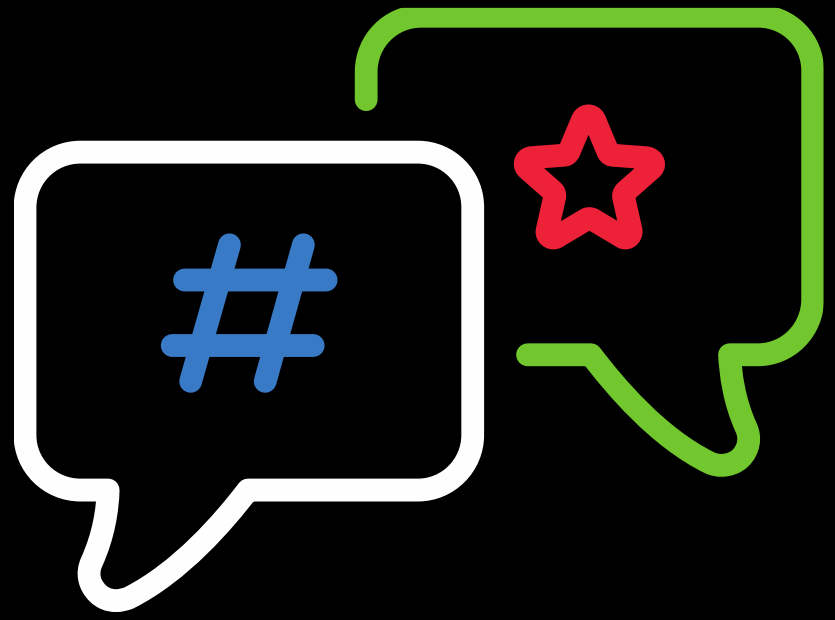
Country	FTE P10	FTE P11	FTE P12	Daily wage	Contributon %	Result account	Working days	Formula	Amount
France	70	69	75	100	20%	Wage	22	FTE P10 * Daily wage * 1	
France	70	69	75	100	20%	Contribution	22	FTE P10 * Daily wage * Contribution %	
France	70	69	75	100	20%	Wage	20	FTE P11 * Daily wage * 1	
France	70	69	75	100	20%	Contribution	20	FTE P11 * Daily wage * Contribution %	
France	70	69	75	100	20%	Wage	18	FTE P12 * Daily wage * 2	
France	70	69	75	100	20%	Contribution	18	FTE P12 * Daily wage * 2 * Contribution %	

Now it’s your turn

Exercise # 5



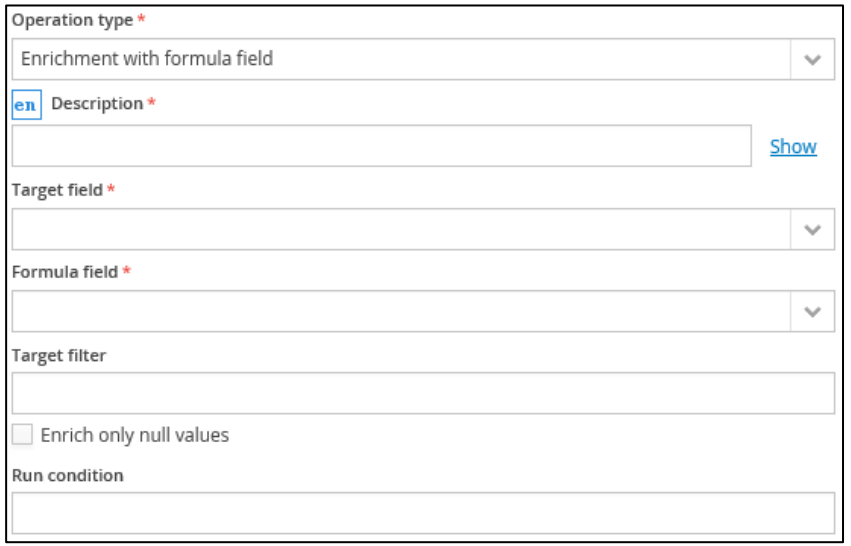
Let’s talk about: Enrichment with formula field



DTP operation – Enrichment with formula field

Enrichment with formula field

- Enrichment of a dataset field
- Through SQL formula
- Written in another dataset field



DTP operations

- DTP applies the formula and stores the result in amount field
- Period for unpivoting is retrieved from a factor of the formula

Country	FTE P10	FTE P11	FTE P12	Daily wage	Contributon %	Result account	Working days	Formula	Amount
France	70	69	75	100	20%	Wage	22	FTE P10 * Daily wage * 1	7000
France	70	69	75	100	20%	Contribution	22	FTE P10 * Daily wage * Contribution %	1400
France	70	69	75	100	20%	Wage	20	FTE P11 * Daily wage * 1	6900
France	70	69	75	100	20%	Contribution	20	FTE P11 * Daily wage * Contribution %	1380
France	70	69	75	100	20%	Wage	18	FTE P12 * Daily wage * 2	15000
France	70	69	75	100	20%	Contribution	18	FTE P12 * Daily wage * 2 * Contribution %	3000

Result

- Dataset structure for reporting purpose

Country	Account	Period	Amount
France	Wage	10	7000
France	Contribution	10	1400
France	Wage	11	6900
France	Contribution	11	1380
France	Wage	12	15000
France	Contribution	12	3000

Delete/group

Consider data deletion or data grouping, when possible, to make data storage more efficient.

Calculation logic in Financial Workspace

Why?

- To book account with result of a calculation algorithm
- High performance tool for complex calculations

Calculation logic type

Sum

- Sum of accounts
- Advanced sums
  - Markup
  - Filters
  - Override
  - Reading from fixed point

Multidimentional logic

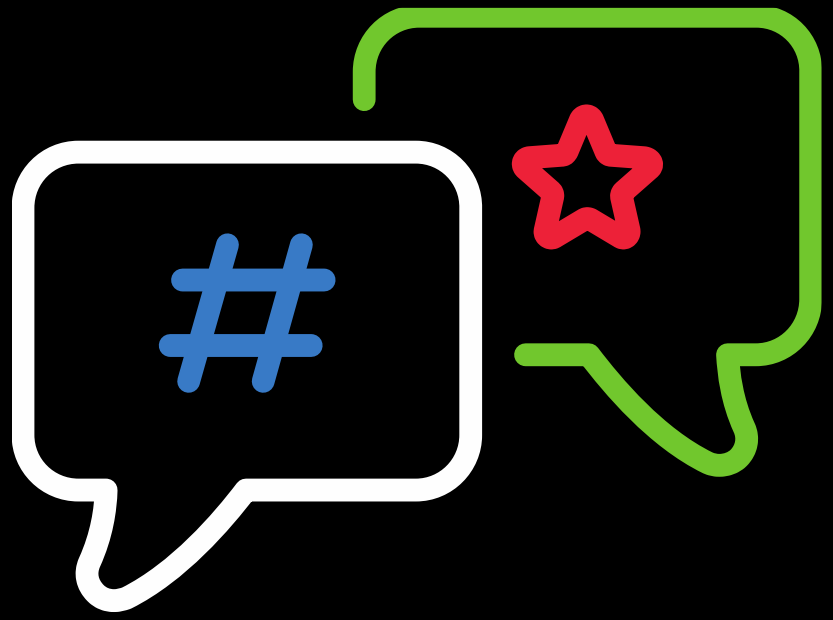
- MDX-Like scripting language (TGKML)
- Used for complex operations
- Reading from different scenario-periods

Now it’s your turn

Exercise # 6



Let’s talk about: Asset Depreciation



Asset depreciation

- Distribution of an asset cost over its expected useful life
- Expected useful life depends on asset class
- Asset cycle accounting
  - Investment: purchase
    - Asset BS account increase
  - Amortization: use
    - Depreciation cost
    - Depreciation fund
  - Disinvestment: sales or disposal
    - Asset BS account decrease
    - Possible gain or loss
- Financial workspace has a built-in calculation for asset depreciation.

5. Asset depreciation

Asset definition

- For each asset class the following is defined:
  - useful life (depreciation periods)
  - PL and BS accounts to be populated

Asset purchase

- For every budgeted purchase the following is defined:
  - asset class
  - purchase date
  - price

Depreciation calculation

- For each period of the current budget the following is calculated:
  - asset investment
  - depreciation
  - depreciation fund

Figures will be available in reporting

5. Asset depreciation assumption

Flat depreciation method

Depreciation starts one month after purchase

Only 3 accounts involved (no disposal, no gain nor loss, no cash flow)

No data from the past

No carry forward to following scenarios

Max useful life: 120 periods (10 years)

Workshops Theory



Reference datasets for

- Configuration definition
- Technical operation

Datasource and virtual dataset to:

- Filter data and perform calculation

Dictionaries to:

- Help configuration selections

5. Asset depreciation – necessary ingredients

Asset class definition

- Reference dataset for class, depreciation periods, accounts
- Reference dataset for period deployments
- Datasource to open rows and calculate percentages
- DTP to store results on a standard dataset

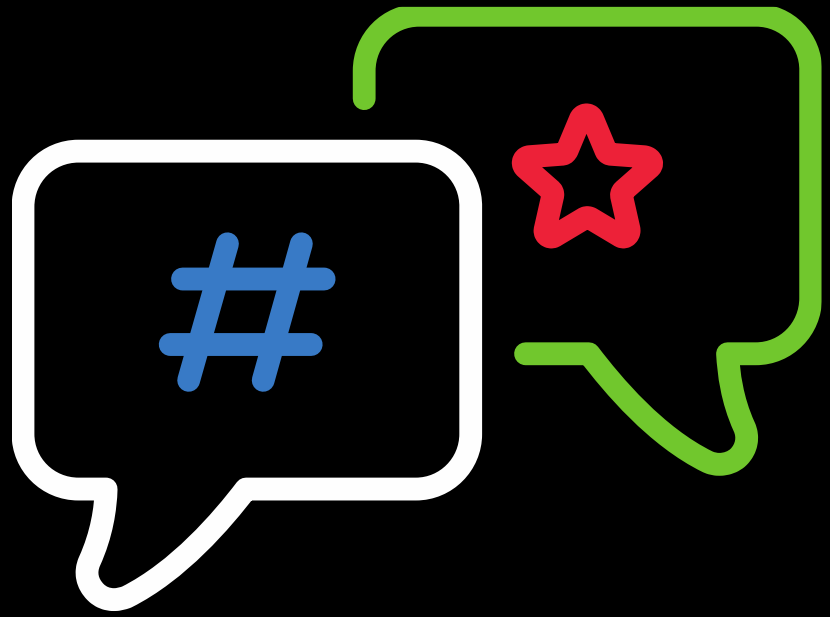
Asset purchase

- Standard dataset to input purchase

Depreciation calculation

- Virtual dataset to join purchase and depreciation
- DTP to filter and map data to store in a reporting format

Let’s talk about: Asset definition



Asset definition

To define asset classes and the relative attributes

Code	Description	Periods of useful life	Asset addition	Depreciation fund	Depreciation cost
Code of the asset class	Description of the asset	Number of depreciation period	Account to store asset purchase value	Account to store the depreciation fund	Account to store the depreciation cost
TA_00	Machinery	36	20020-020 Machinery addition	20020-080 Machinery depreciation fund	50370 Machinery depreciation
…	…	…	…	…	…

How to move from definition to a table with open rows populated with flat percentages?

- This format makes the next calculations easier

Definition

ASSET CLASS	PERIODS OF USERFUL LIFE
Machinery	50
Equipment	20
Building	120
…	…

Prepopulation

ASSET CLASS	PERIOD	DEPRECIATION
Machinery	1	2%
Machinery	2	2%
Machinery	3	2%
Machinery	4	2%
…	…	…
Equipment	1	5%
Equipment	2	5%
Equipment	3	5%
…	…	…

Flat percentage calculation

- Rows can be opened with a datasource with asset definition cross joining period deployment
- Filtering only where period <= period of useful life

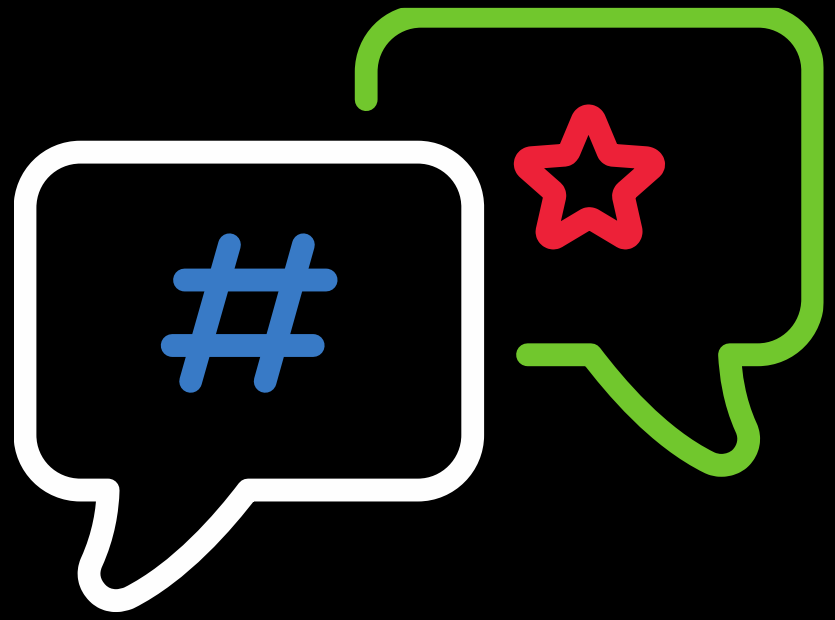
Asset class	Period of useful life	Period	Depreciation
From definition	From definition	From period deployment	Calculated 1/useful life
Equipment	20	1	5%
Equipment	20	2	5%
Equipment	20	…	5%
Equipment	20	20	5%
Equipment	20	21	…
Equipment	20	…	…
Equipment	20	120	…

Now it’s your turn

Exercise # 7



Let’s talk about: Asset depreciation calculation



5. Asset depreciation – necessary ingredients

Asset class definition

- Reference dataset for class, depreciation periods, accounts
- Reference dataset for period deployments
- Datasource to open rows and calculate percentages
- DTP to store results on a standard dataset

Asset purchase

- Standard dataset to input purchase

Depreciation calculation

- Virtual dataset to join purchase and depreciation
- DTP to filter and map data to store in a reporting format

Dataset
60 Asset purchase

Partitioning

- by business cycle (process)

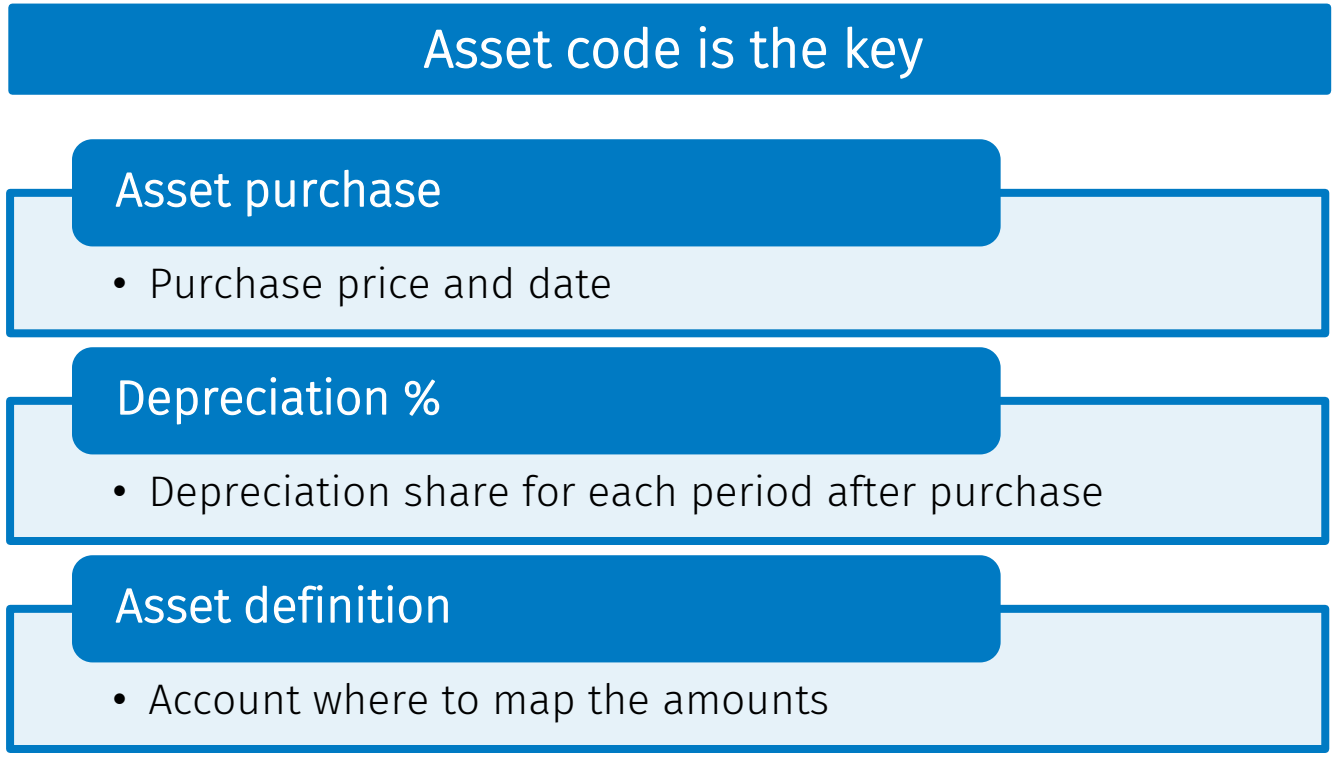
Fields

- Entity
- Asset name-code-description
- Year and month of purchase
- Price

Time dimensions specified as year-month

- Makes calculations easier
- Year-month are text fields
  - Question: how to filter data in form?
  - Answer: with reporting with custom filters

Depreciation sources – Virtual dataset



Depreciation calculation – Virtual dataset

To calculate

Depreciation amount of each period

- Price * depreciation %

Calendar period of depreciation

- Purchase month + period of depreciation

Depreciation calculation

Virtual dataset fields to prepare data for depreciation calculation

Asset code	Asset desc	Asset name	Year	Month	Price	Depr %	Depr period	Acc. Add.	Acc. Depr. Fund.	Acc. Depr.
Key	from definition	from purchase	from purchase	from purchase	from purchase	from depr. def. result	from depr. def. result	from definition	from definition	from definition
TA-00	Equipment	PC	2058	9	2000	20%	1	20020	20080	50370
TA-00	Equipment	PC	2058	9	2000	20%	2	20020	20080	50370
TA-00	Equipment	PC	2058	9	2000	20%	3	20020	20080	50370
TA-00	Equipment	PC	2058	9	2000	20%	4	20020	20080	50370
TA-00	Equipment	PC	2058	9	2000	20%	5	20020	20080	50370

Depreciation calculation

Virtual dataset formulas to prepare data for depreciation calculation

Asset code	Asset desc	Asset name	Year	Month	Price	Depr %	Depr period	Period	Amount
Key	from definition	from purchase	from purchase	from purchase	from purchase	from depr. def. result	from depr. def. result	Month + Depr period	Price * Depr %
TA-00	Equipment	PC	2058	9	2000	20%	1	10	400
TA-00	Equipment	PC	2058	9	2000	20%	2	11	400
TA-00	Equipment	PC	2058	9	2000	20%	3	12	400
TA-00	Equipment	PC	2058	9	2000	20%	4	13	400
TA-00	Equipment	PC	2058	9	2000	20%	5	14	400

Now it’s your turn

Exercise # 8



Depreciation calculation and storage – DTP

Result is stored on the 3 accounts specified for:

Asset investment

- Filter to input price on purchase month
- Scenario retrieved from context
- Transformation to write period in a suitable way

Depreciation fund

- Filter to input only period within the planning year
- Change sign to amount
- Scenario and period as above

Depreciation

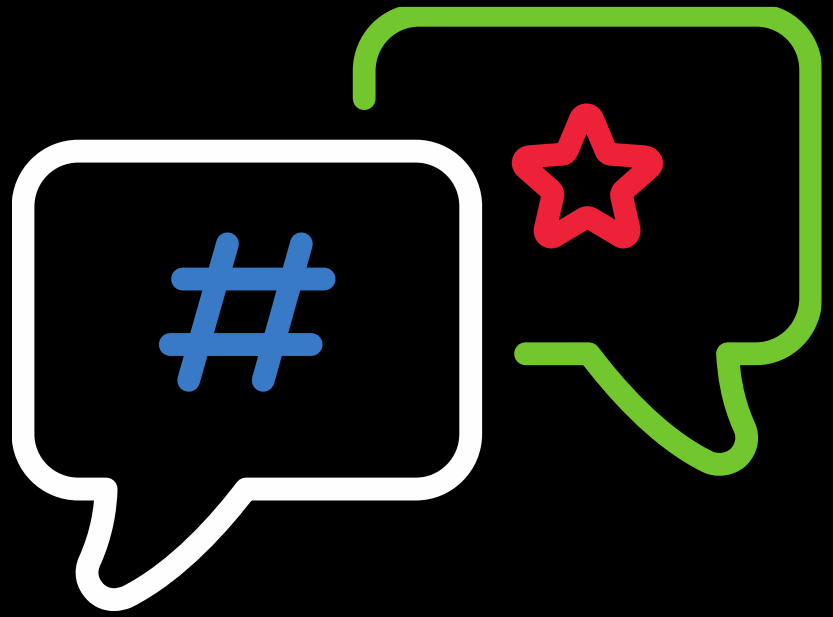
- Filter to input only period within the planning year
- Scenario and period as above

Now it’s your turn

Exercise # 9



Let’s talk about: Reporting



Reporting

Data are moved to the different data entry and calculation structures to a standard structure for reporting

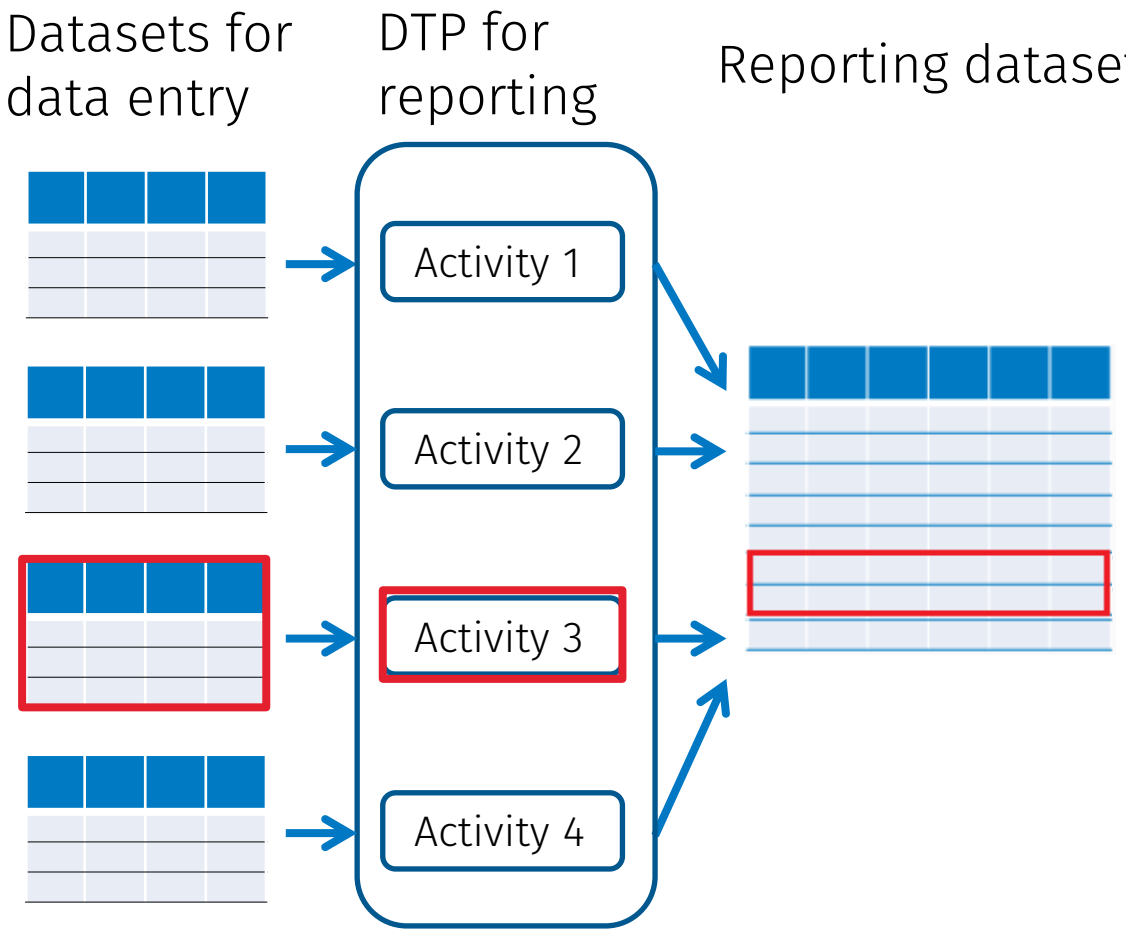
- All dimensions + amount should be present in reporting structure to leverage standard built-in functionalities (hierarchies, filters, parameters, time rollup, conversion…)
- 1 or more DTP are triggered to update reporting dataset



Pay attention to:

- execution condition -> avoid running unnecessary data movements
- deletion policy -> reporting layer filter may be needed

Reporting DTP – Execution condition and deletion policy



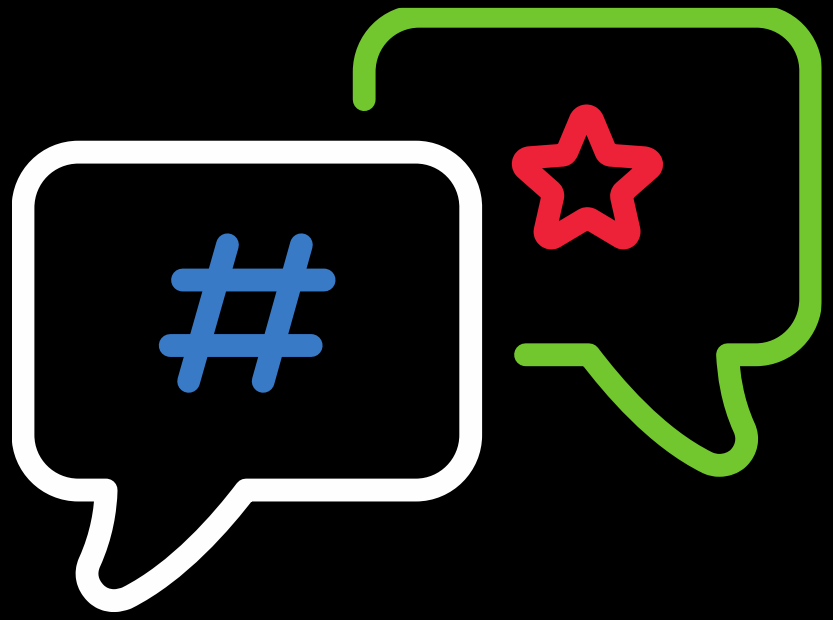
- Run condition allows you to run only the activity relative to a data entry
- Reporting layer filter allows you to delete from the target only data managed by the activity

Now it’s your turn

Exercise # 10

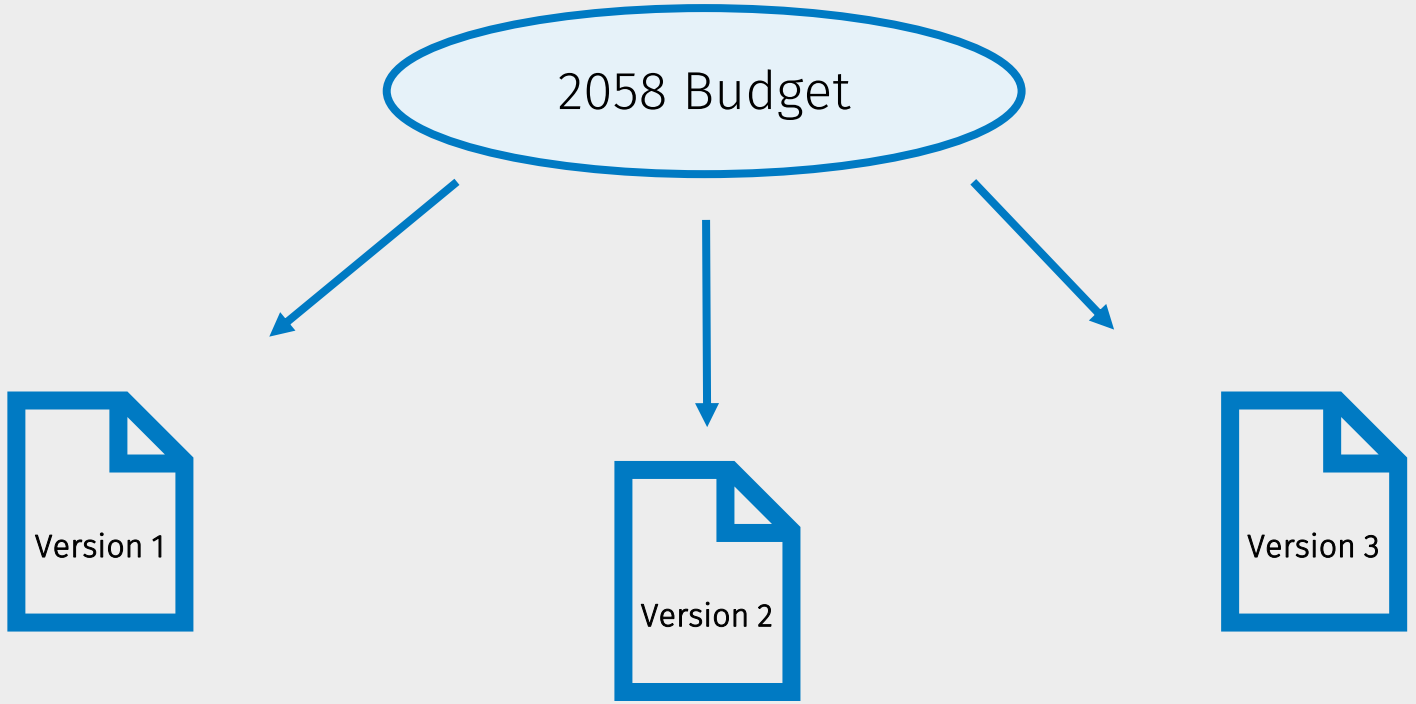


Let’s talk about: Versioning: Snapshot scenarios



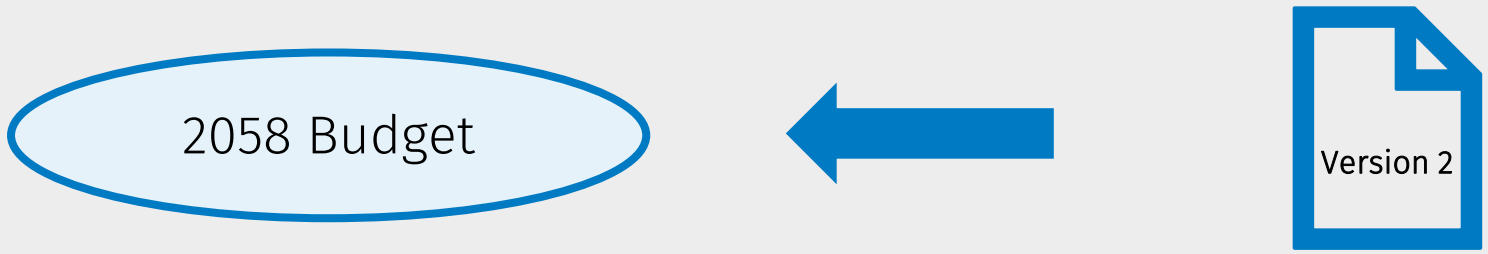
Snapshot scenarios

- Used to keep several versions of planning data related to the same scenario
- User can create different Snapshot scenarios where data will be copied and frozen



Snapshot scenarios

- Different versions will be compared and user can choose the preferred one and restore it on original scenario



Snapshot configuration

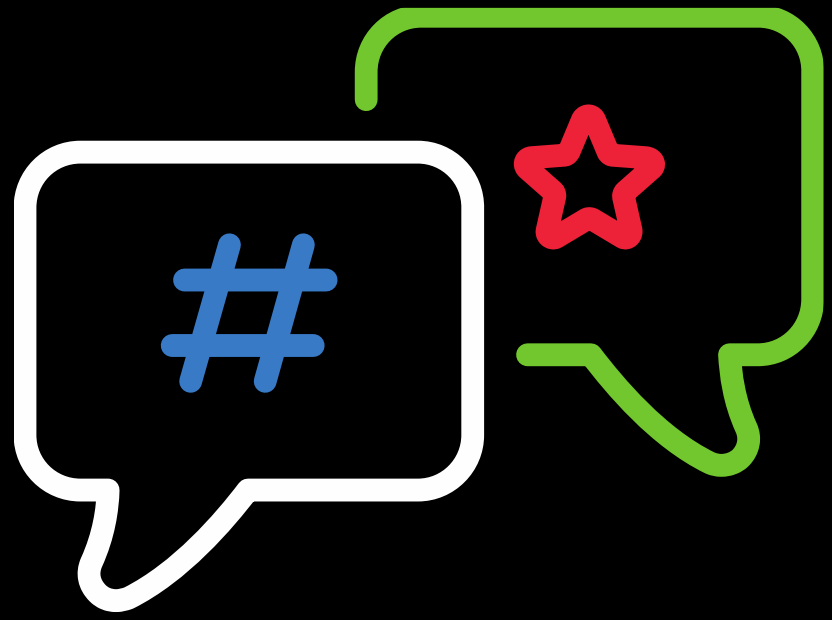
- Scenarios with the same periods opened must be linked to original scenario
- Snapshot data processing must be enabled in the process configuration
  - Run by:
    - Process
    - Scenario
    - Scenario/period
  - Filter on dimensions
    - Entity
    - Category
    - Custom dimensions
- Creation of snapshot and restoring data from snapshot scenario can be run from process cockpit actions
- The snapshot filters and copies the dataset measures based on the process step

Now it’s your turn

Exercise # 11



Let’s talk about: Forecasting



Forecasting – the what

The budget is essentially a roadmap showing where a company aims to go

The forecast indicates where a business is actually heading

Forecast helps and updates budgeting & planning, estimating for the future.
Based on past events and management insight.

Forecasting – the why

Static budgets are not always fit for purpose

Budgets will often be outdated by the time they are finished

Instead, forecasting updates planning, allowing companies to make better informed decisions.

Forecasting fosters agility in competitive and variable business landscapes.

Judgement vs quantitative forecasting

Combining judgment and quantitative forecasting gets the best results

Judgement forecasting

- Uses intuition and experience to make connections and understand the context
- Risk of bias and difficulties to analyse large volumes of data
- Best when there is little historical data (e.g. new product, new plan, competitor actions)

Quantitative forecasting

- Uses analytics to analyse data to discern trends and patterns.
- Risky when there is little data or in the analysis of new contexts
- Best when data shows repeated patterns

Statistical forecasting

Time series forecasting

- **Purpose**: tactical, short-term, determining seasonal patterns
- **Requires**: historical data
- **Results**: short-term adjustments
- **Applications**: demand planning, production planning, inventory management

- Predictive intelligence

Model based forecasting

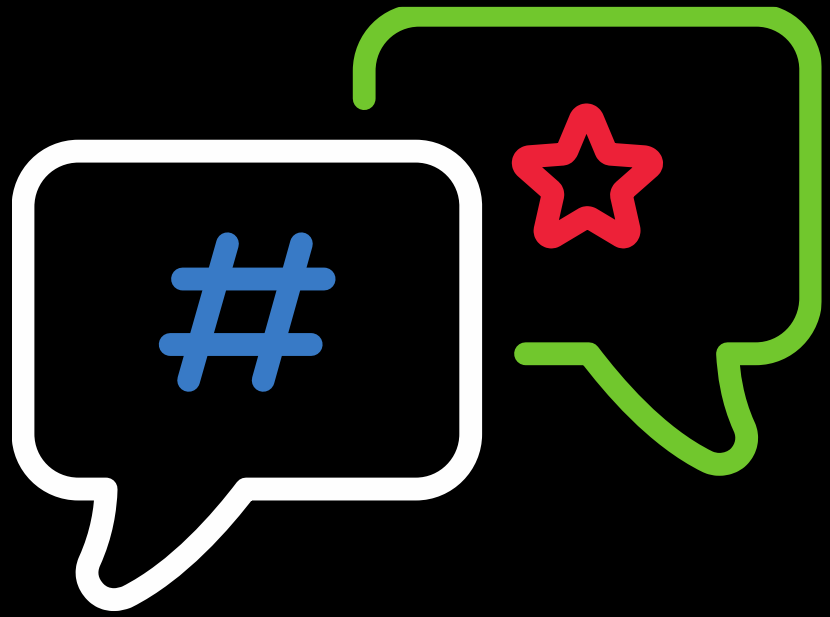
- **Purpose**: strategic, accounting for changes in business environment
- **Requires**: management insight
- **Results**: change in business practice
- **Applications**: marketing, resource allocation, business optimization, comparisons of business plans

- Forecast models and similar

Forecasting in CCH Tagetik

- Connect planning, budget and actual data to provide the forecast for the required purpose
- Analytic Information Hub allows you to manage different data models suitable for any function involved in forecasting process.
- Predictive intelligence provides estimates based on machine learning
- Financial workspace provides built-in forecast models for financial forecasting

Let’s talk about: Forecast models



Forecast models - Approaches

Time Horizon

- Static (3+9, 6+6, 9+3)
- Rolling (12,18,24,..)

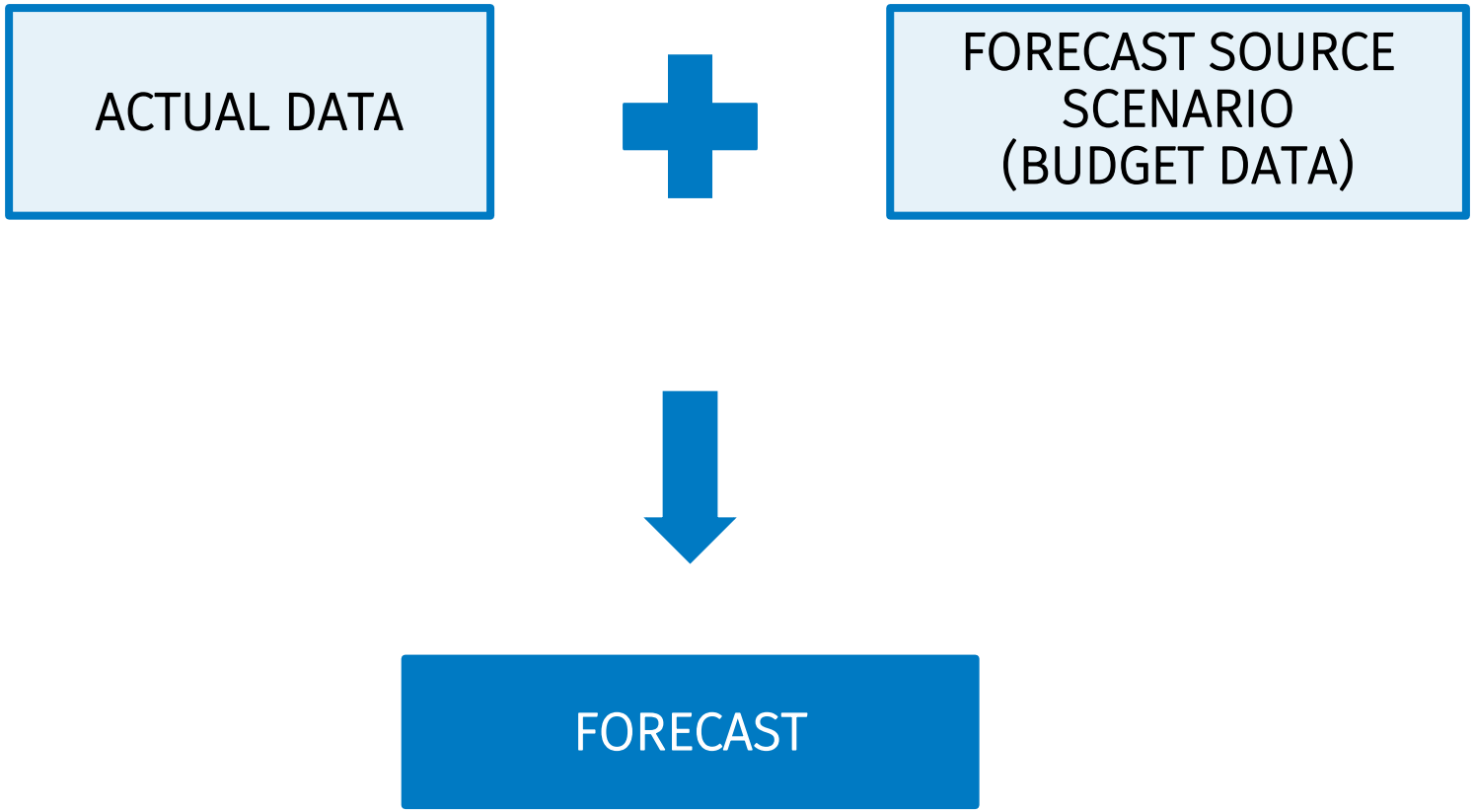
Data Granularity

- Aggregated
- Very Detailed (≈ Budget)

Methodology

- Zero Based
- Automatically Pre-Seeded

Forecast Model



Forecast Model logic: Hold the to-year

MODEL TYPE: Hold the to-year

Copy the existing Actual months onto Forecast scenario

Complete the missing periods with Budget data (or any suitable source scenario)

		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Input	Actuals	67	87	88	106	89	93							**530**
Input	Budget or previous Forecast	100	90	80	70	100	100	60	70	250	90	100	110	**1.220**
Output	Current forecast	67	87	88	106	89	93	60	70	250	90	100	110	**1.210**

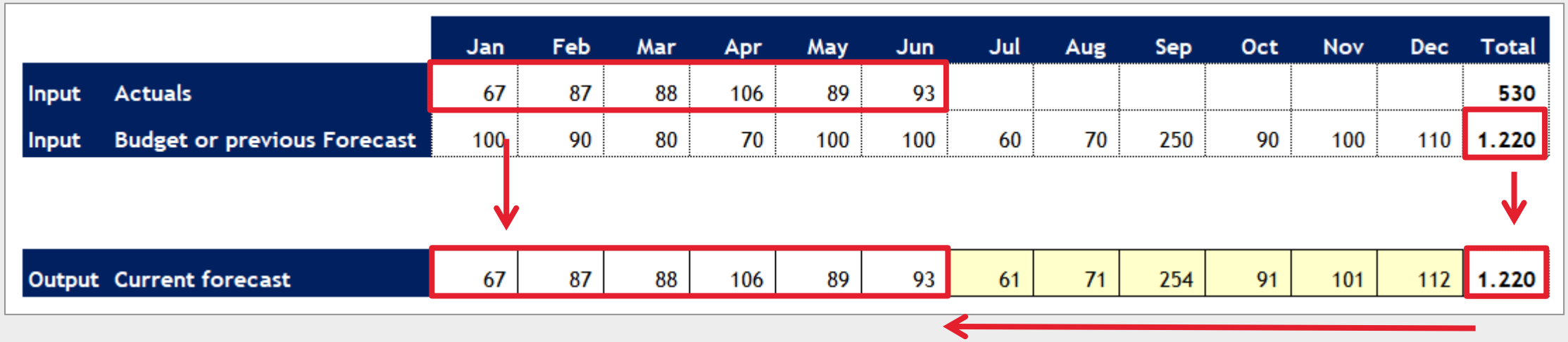
Forecast Model: Hold the annual total

MODEL TYPE: Hold the annual total

Copy the existing Actual months onto Forecast scenario

Set the Forecast annual total equal to Budget annual total

Populate missing periods spreading the difference between Actual and annual total

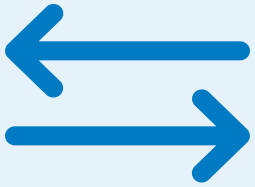


		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Input	Actuals	67	87	88	106	89	93							530
Input	Budget or previous Forecast	100	90	80	70	100	100	60	70	250	90	100	110	1.220
Output	Current forecast	67	87	88	106	89	93	61	71	254	91	101	112	1.220

About this approach



Forecast is created just by copying data.



Once copied, data is likely to be modified.



Common approach among many customers.
It can be modified to meet specific needs.

Example – Static Forecasting

Assumptions:

- Fiscal year is 2020 and forecasting month is 5
- All data is forecasted monthly using hold the to-year
- Budget is used as forecast source for months 2-3-4
- Previous scenario is used as forecast source for month 5

		01	02	03	04	05	06	07	08	09	10	11	12	Total
Actual	2020ACT	97	80	100	96									**373**

		01	02	03	04	05	06	07	08	09	10	11	12	Total
Budget	2020BDG	87	62	55	92	57	96	76	79	61	90	74	82	**911**

		01	02	03	04	05	06	07	08	09	10	11	12	Total
Initial Copy	2020FCT_02	97	62	55	92	57	96	76	79	61	90	74	82	**921**
Initial Copy	2020FCT_03	97	80	55	92	57	96	76	79	61	90	74	82	**939**
Initial Copy	2020FCT_04	97	80	100	92	57	96	76	79	61	90	74	82	**984**
Modification	2020FCT_04	97	80	100	101	63	106	84	87	67	99	81	90	**1055**
Initial Copy	2020FCT_05	97	80	100	96	63	106	84	87	67	99	81	90	**1050**
Modification	2020FCT_05	97	80	100	96	69	116	92	96	74	109	90	99	**1117**

Key Takeaways:

- By Forecasting Monthly we are storing a history of the forecast
- This model gives us flexibility to pre-populate new forecasts based on different scenarios

Forecasting workflow

- Copy data on a forecast scenario selecting:
  - Actual data for until available
  - 1st forecast period
  - Forecast source scenario
- The forecasted periods must be populated according to the forecast model selected when launching the copy tool

Workshops Theory



Datasource and datasetpartition function to:

- Retrieve data from the required partitions

Reporting parameters to:

- Select and filter dynamically the information needed for the data processing

Run condition to:

- Execute only the data processing needed for the model selected

4. Personnel planning – necessary ingredients

Source data

- 2 datasources to retrieve actual and forecast source data dynamically

Parameters

- Actual scenario
- First forecast period
- Forecast source scenario
- Forecast model

Copy data and forecast calculations

- DTP with operation executed according to the model selected
- Standard dataset to store results

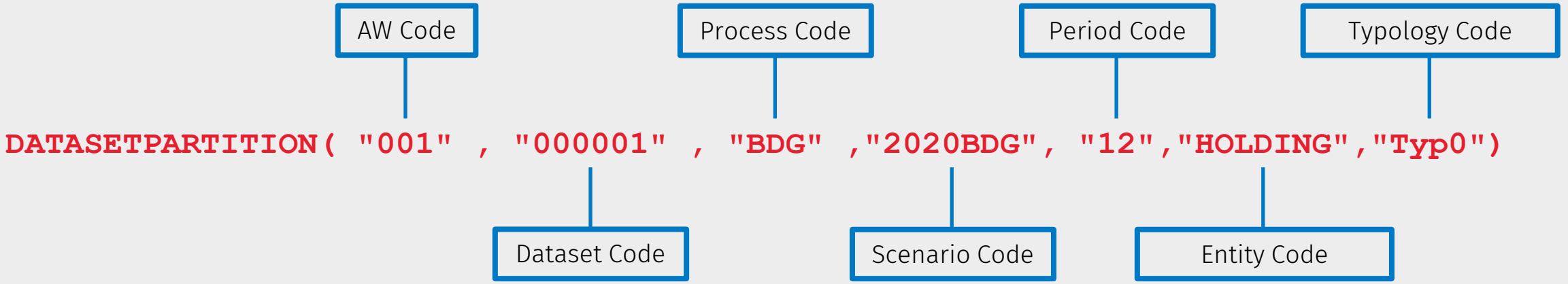
Forecasting – more possibilities (not in the exercise)

- **Mark up**: to apply to forecast source data
  - numeric coefficient -> parameter
- **Aggregation**: to reduce the level of detail of actual and forecast data
  - lookup table -> reference dataset or hierarchy
- **Filters**: to process only a subset of data:
  - hierarchy node -> parameter
- **Deletion**: to avoid deletion on target scenario:
  - rule -> reporting layer filter
- **More models**: to have more calculation options:
  - More selections and operations -> run condition

Physical name of a partition (recap)

Function - DATASETPARTITION ():

To retrieve the name of a dataset partition table based on its names and context.



PROCESS($Scenario.code,$Period.code).code → Useful to retrieve Process code

Parameters and Run condition (recap)

Dimensional attribute

- ${$Scenario.prev.code}

Reporting parameter

- ${@PARAMETER}

Run condition

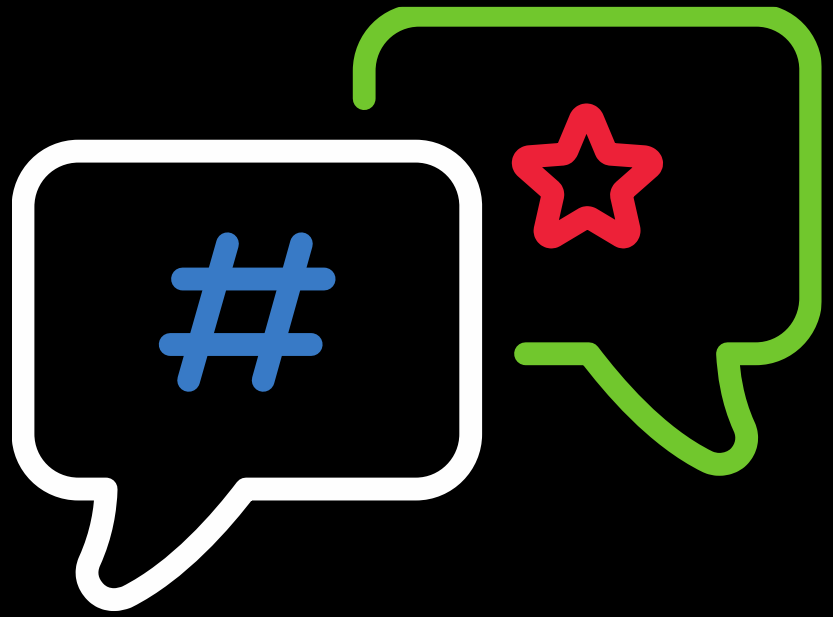
- to execute the activity or not

Now it’s your turn

Exercise # 12



Let’s talk about: Rolling Forecast



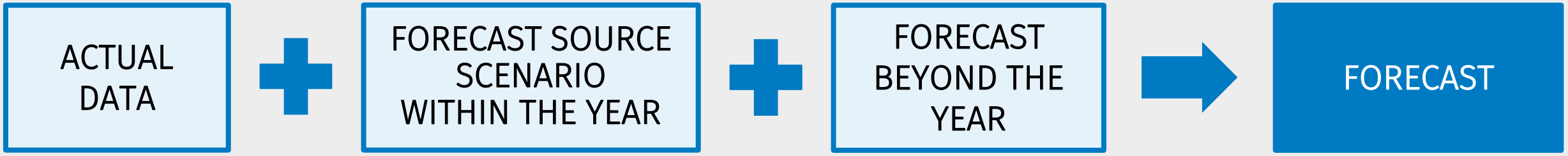
From static to rolling forecast

Static forecasts project numbers on a single fixed time frame (say, till year end)

Rolling forecasts drop a month as it passes, forecasting the next month automatically

Planning continuously over a predetermined time horizon. Based on the most recent numbers and time frame.

Rolling forecast - how



Actual scenario for the periods available

Current year Forecast scenario can be prefilled by Forecast model

The challenge is how to prefill the following year Forecast scenario?

Rolling forecast – scenario management

12 Month Rolling Forecast

Similar setup, with more scenarios involved

2021RF_02_20

Year of forecasted data

Month in which we are forecasting

Year in which we are forecasting

	01	02	03	04	05	06	07	08	09	10	11	12	2020 Total
2020ACT	97	80											**177**

	01	02	03	04	05	06	07	08	09	10	11	12	2020 Total
2020BDG	87	62	55	92	57	96	76	79	61	90	74	82	**911**

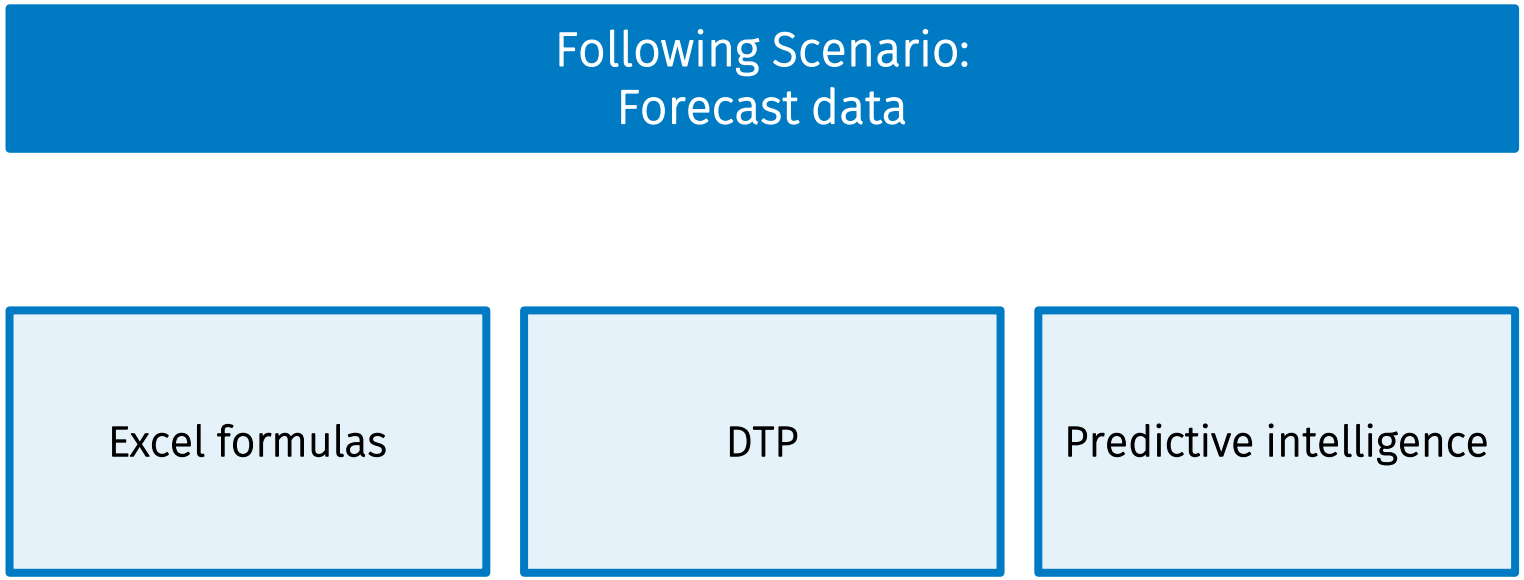
	01	02	03	04	05	06	07	08	09	10	11	12	2020 Total
2020RF_02_20	97	62	55	92	57	96	76	79	61	90	74	82	**921**
2020RF_03_20	97	80	55	92	57	96	76	79	61	90	74	82	**939**

	01	02	...
2021RF_02_20	65		
2021RF_03_20	65	76	

Static Scenarios: Forecast model can be used

Rolling Forecast Scenarios: Data entry of calculated

Fill of following forecast scenario





Takeaway

Join CCH Tagetik Insiders Hub!

Get access to exclusive CCH Tagetik best practices and tips, sneak peaks at product updates

Share the love by participating in Testimonials, Case studies, Surveys, Reviews, Referrals

Leverage your social capital by connecting with other finance and operations professionals

Earn points up the leader board and redeem rewards

Here's the way the hub works

1 Log into the CCH Tagetik Community and click on the Insiders tab

2 Poke around and complete challenges earning points to redeem rewards

3 Get recognized for all the support you give CCH Tagetik

It's that easy!

Join CCH Tagetik Insiders now!

What other learning assets are available?

User Manual

CCH Tagetik Community

Knowledge Base

What's next?

Visit the Academy webpage

Next training and/or Assessment

Mail to info.academy@wolterskluwer.com

Thank you



Wolters Kluwer

CCH® Tagetik | ACADEMY